HIGH DYNAMIC RANGE IMAGING

Computational Imaging – Spring 2024

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Slide credits

Many of these slides are based on the following content:

- Computational Photography (15-463). Carnegie Mellon University.
- Computational Photography. Brown University.

Which in turn were inspired by or adapted from:

- James Hays (Georgia Tech).
- Fredo Durand (MIT).
- Gordon Wetzstein (Stanford).
- Marc Levoy (Stanford, Google).
- Sylvain Paris (Adobe).
- Sam Hasinoff (Google).

Plenoptic imaging

Basis:

Digital photography (Lecture #1)
Mathematical foundations (Lecture #2)

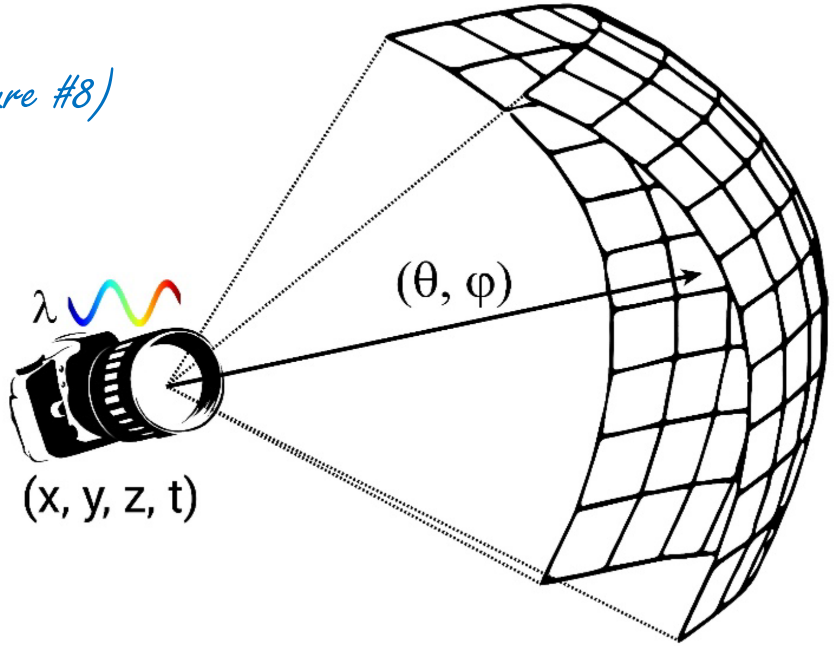
Transient imaging (Lecture #8)

Hyperspectral imaging (Lecture #6)

$$P = P(\theta, \phi, \lambda, t, x, y, z)$$

Light field imaging (Lecture #4)

High dynamic range imaging (Lecture #5)



Plus common tools and techniques:

Coded photography (Lecture #3)
Compressive sensing (Lecture #3)
Computational illumination (Lecture #7)

Remember from Lecture 1...

Photodiode response

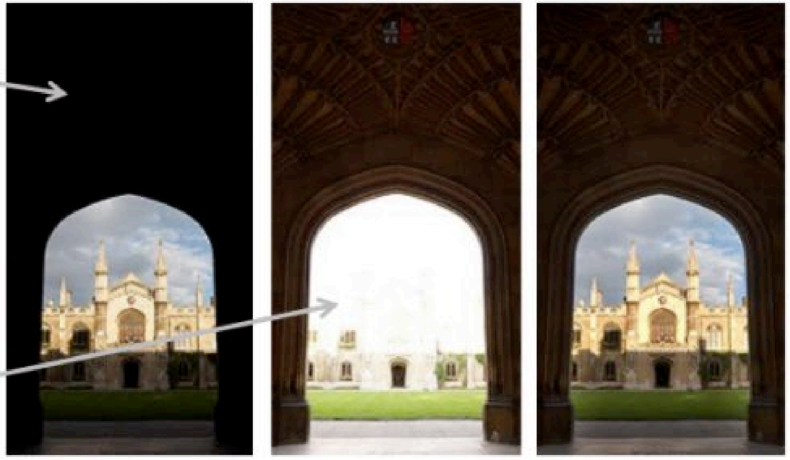
The photodiode response is <u>mostly</u> linear, but:

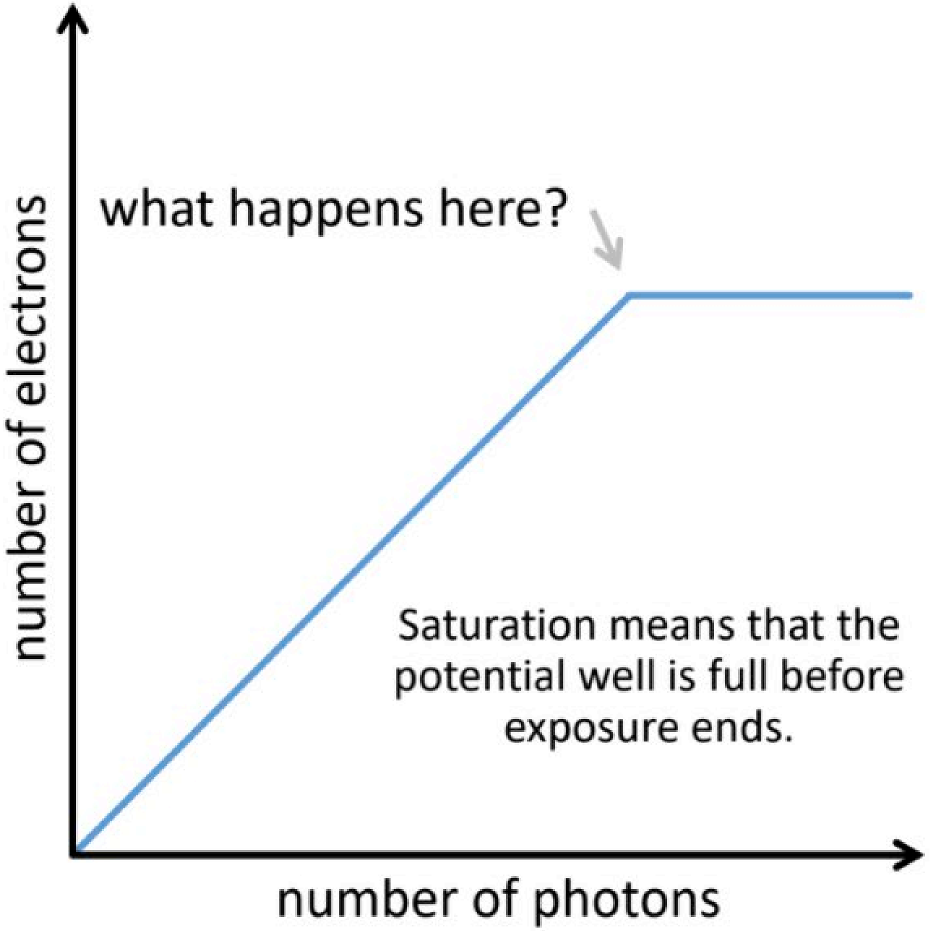
- non-linear when potential well is saturated (over-exposure)
- non-linear near zero (due to noise)

High-dynamic-range imaging: Later in the course!

under-exposure (non-linearity due to sensor noise)

over-exposure (non-linearity due to sensor saturation)



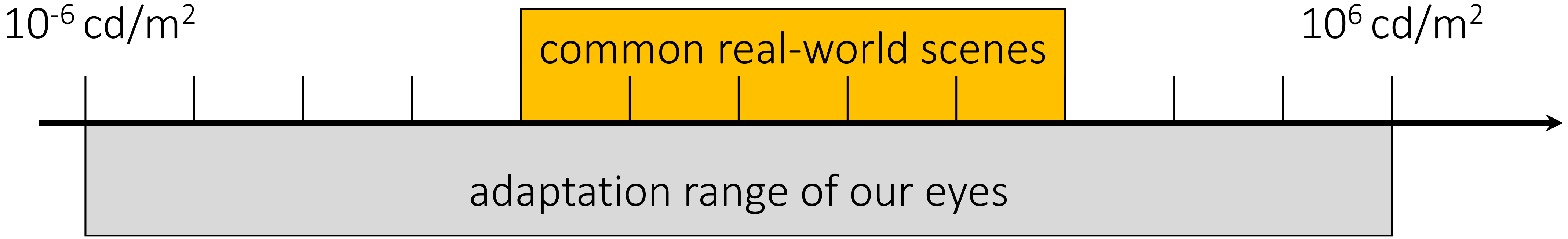


The problem of dynamic range

Dynamic range: Ratio between the largest and smallest values of luminance that a given imaging system can capture or display.

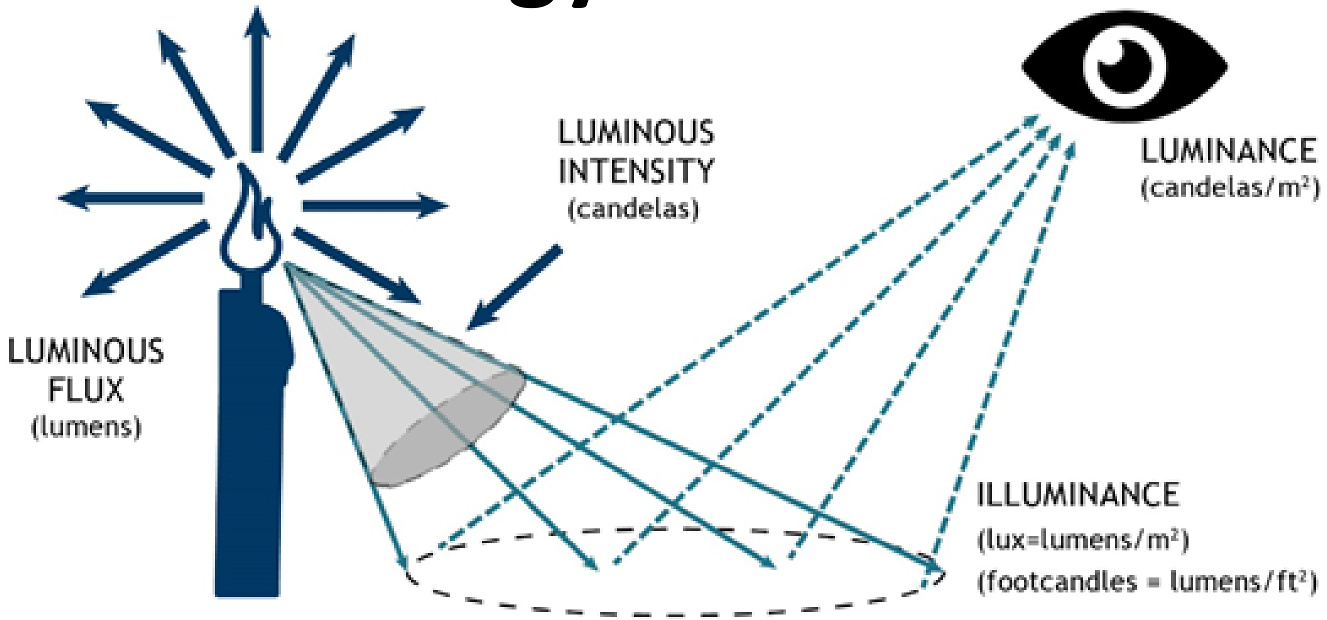
The problem of dynamic range

- The world has a high dynamic range



Sources: [CMU 15-463]

A note about terminology



Luminance: Amount of light emitted from or reflected off surface in a specific direction
Illuminance: Amount of light that shines onto a surface (how much light is available to be seen by the human eye)
Luminous flux: Amount of light a light source emits, integrated over the entire angular span of the light (weighted by human's eye sensitivity to different wavelengths)
Luminous intensity: Measure of the light that shines from the source in a given direction

Sources: [CMU 15-463]

A note about terminology

Photometric Units		Radiometric Units	
Luminous flux	**lm**	Radiant flux	**W**
Luminous intensity	cd† = lm/sr⋆	Radiant intensity	W/sr
Illuminance	lux = lm/m^2	Irradiance	W/m^2
Luminance	$cd/m^2 = lm/sr/m^2$	Radiance	$W/sr/m^2$

† cd refers to candelas; ⋆ sr refers to steradian.

Photometry: Measurement of light detectable by the human eye

Radiometry: Measurement of optical radiation (in the frequency range between 3×10^{11} Hz and 3×10^{16} Hz)

Sources: [CMU 15-463]

A note about terminology

From Luminous Flux to Illuminance:

$$\text{Illuminance}(E) = \frac{\text{Luminous Flux}(\Phi)}{\text{Area}(A)}$$

From Luminous Flux to Luminous Intensity:

$$\text{Luminous Intensity}(I) = \frac{\text{Luminous Flux}(\Phi)}{\text{Solid Angle}(\Omega)}$$

Relating Luminous Intensity to Illuminance:

$$E = \frac{I}{r^2}$$

From Luminance to Luminous Intensity:

$$\text{Luminous Intensity}(I) = \text{Luminance}(L) \times \text{Area}(A) \times \text{Solid Angle}(\Omega)$$

The problem of dynamic range

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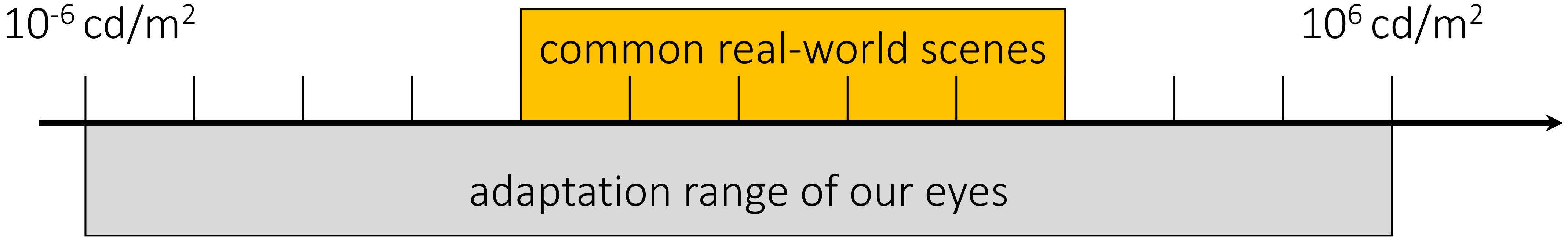
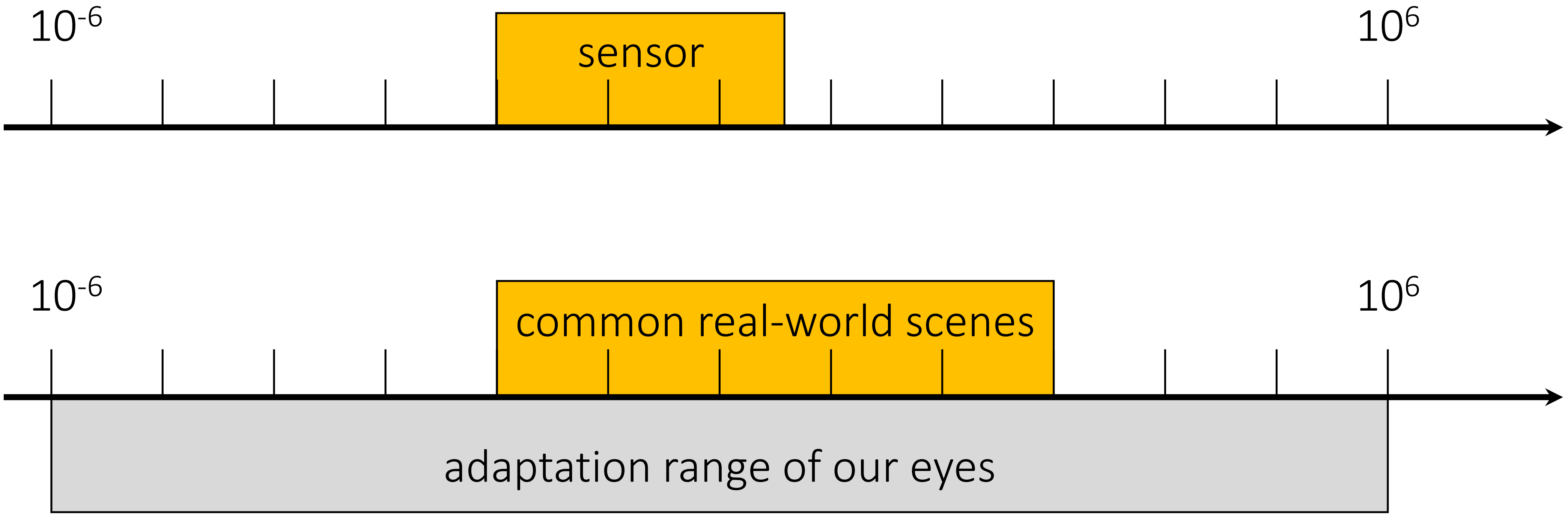


The problem of dynamic range

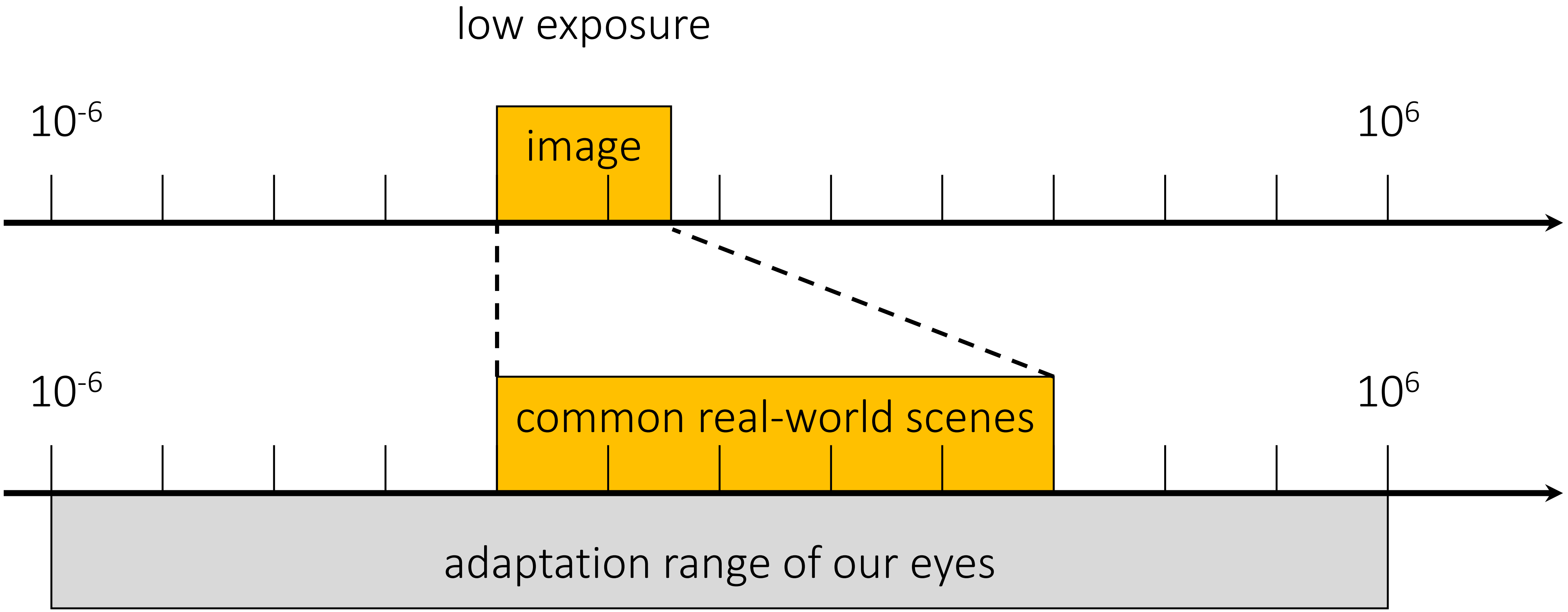
- Digital sensors have a low dynamic range



Sources: CMU 15-463

The problem of dynamic range

- Digital images have an even lower dynamic range



Sources: [CMU 15-463]

The problem of dynamic range

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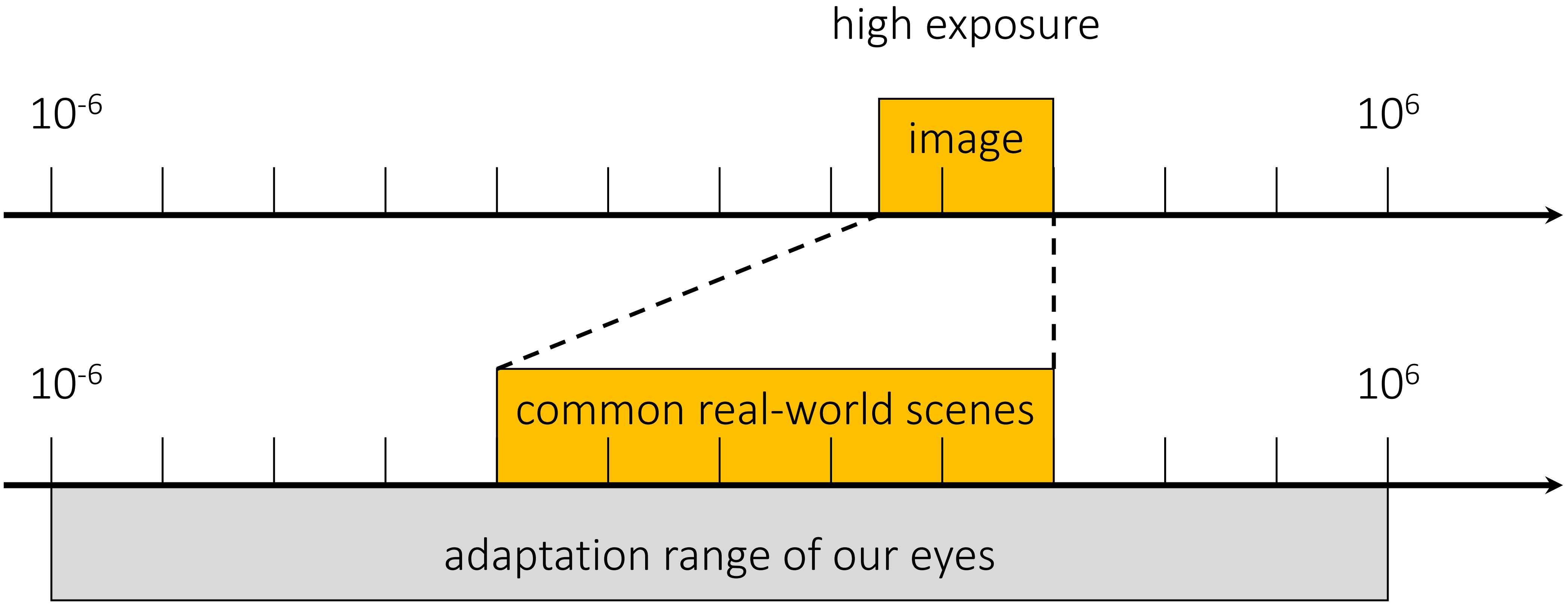

The problem of dynamic range

- Our devices do not match the real world

• 10:1	photographic print (higher for glossy paper)
• 200:1	slide film
• 500:1	negative film
• 1,000:1	LCD display
• 2,000:1	digital SLR (at 12 bits)
• 100,000:1	real world

Sources: CMU 15-463

The problem of dynamic range

- Our devices do not match the real world

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- 200:1 slide film
- 500:1 negative film
- 1,000:1 LCD display
- 2,000:1 digital SLR (at 12 bits)
- 100,000:1 real world

Two challenges:

1. *HDR imaging: Which parts of the world do we include in the 8-14 bits available to our capture device?*
2. *Tonemapping: Which parts of the world do we display in the 4-10 bits available to our display device?*

The problem of dynamic range

- Our devices do not match the real world

- 10:1 photographic print (higher for glossy paper)
- 200:1 slide film
- 500:1 negative film
- 1,000:1 LCD display
- 2,000:1 digital SLR (at 12 bits)
- 100,000:1 real world

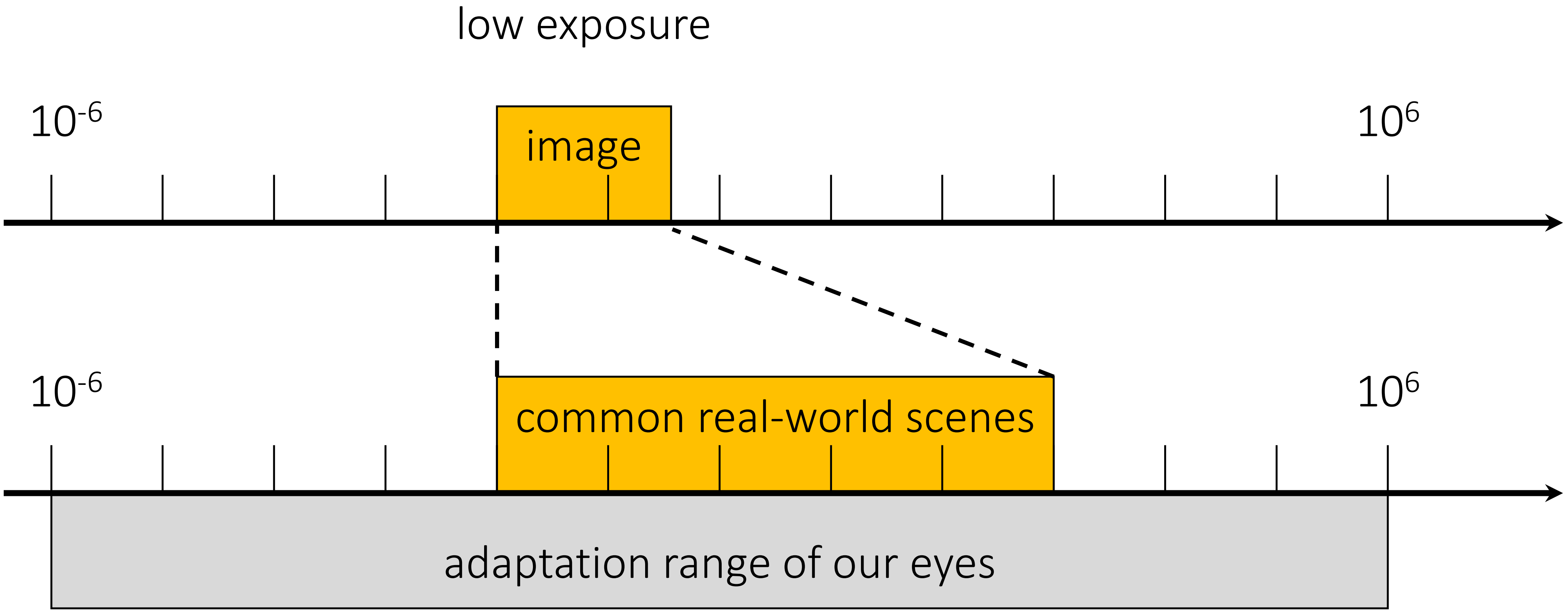
~~Two~~ Three challenges:

1. HDR imaging: Which parts of the world do we include in the 8-14 bits available to our capture device?
2. Tonemapping: Which parts of the world do we display in the 4-10 bits available to our display device?
3. Inverse tonemapping: How do we show conventional images in HDR displays?

Light metering

Light metering

- When encountered with this problem, what does our conventional camera do?



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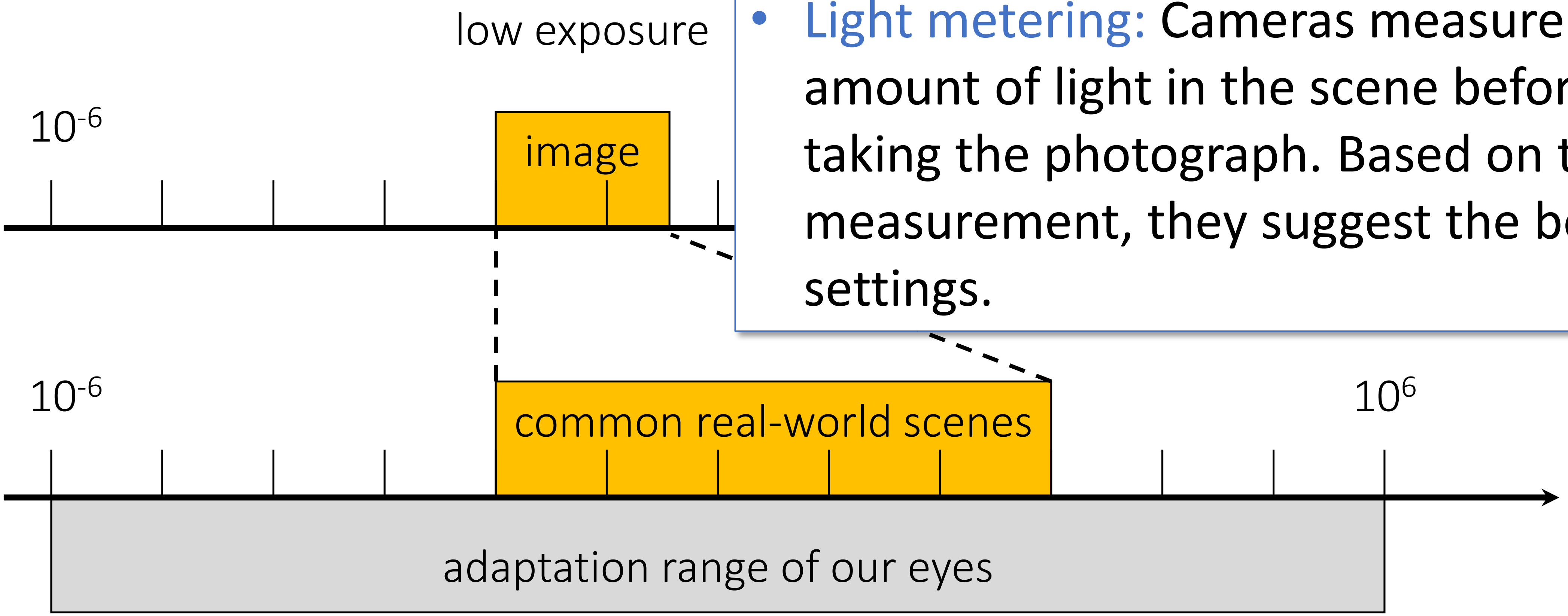


- Light metering: Cameras measure the amount of light in the scene before taking the photograph. Based on this measurement, they suggest the best settings.

Light metering

Aperture priority ("A"): you set aperture, camera sets everything else.

- Pros: Direct depth of field control.
- Cons: Can require impossible shutter speed (e.g. with f/1.4 for a bright scene).

Shutter speed priority ("S"): you set shutter speed, camera sets everything else.

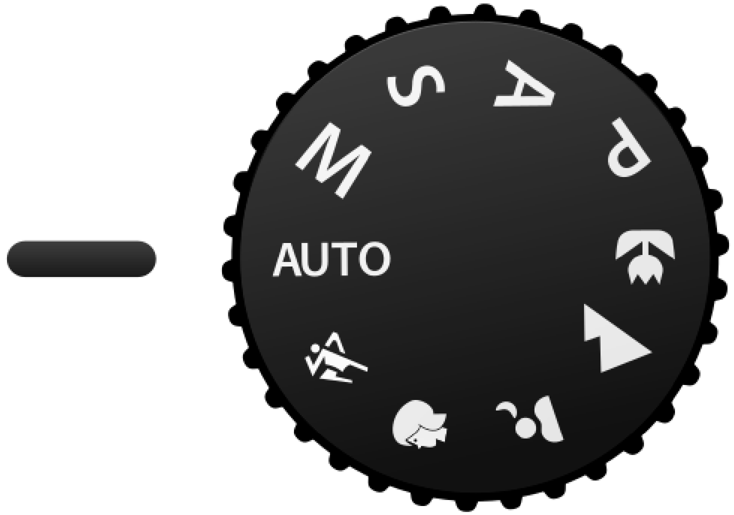
- Pros: Direct motion blur control.
- Cons: Can require impossible aperture (e.g. when requesting a 1/1000 speed for a dark scene)

Automatic ("AUTO"): camera sets everything.

- Pros: Very fast, requires no experience.
- Cons: No control.

Manual ("M"): you set everything.

- Pros: Full control.
- Cons: Very slow, requires a lot of experience.



generic camera mode dial

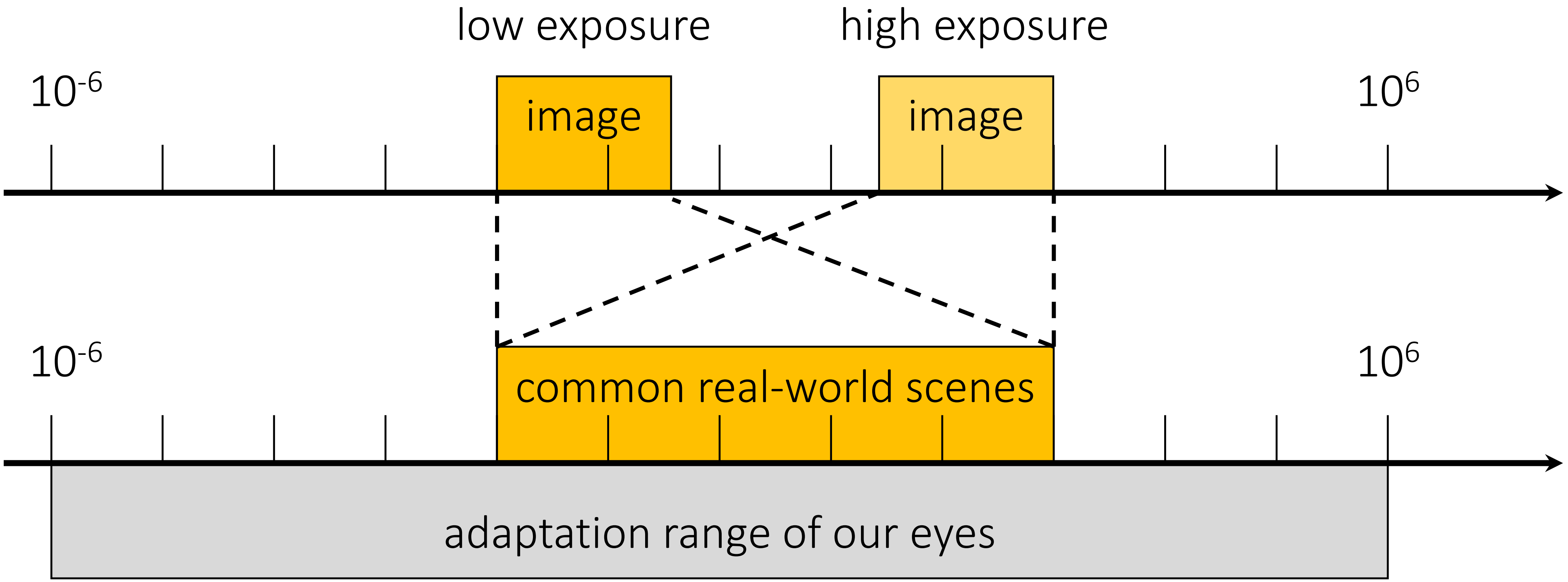
High dynamic range imaging (HDRI)

- Multibracketing
- Merging w/ RAW images
- Merging w/ non-linear images (radiometric calibration)
- Other methods for HDR imaging

Sources: [CMU 15-463]

High dynamic range imaging

- We can adjust the exposure given a certain scene




HYATT


HYATT


HYATT

High dynamic range imaging

- But none of these images contains all the relevant information, each contains part of it

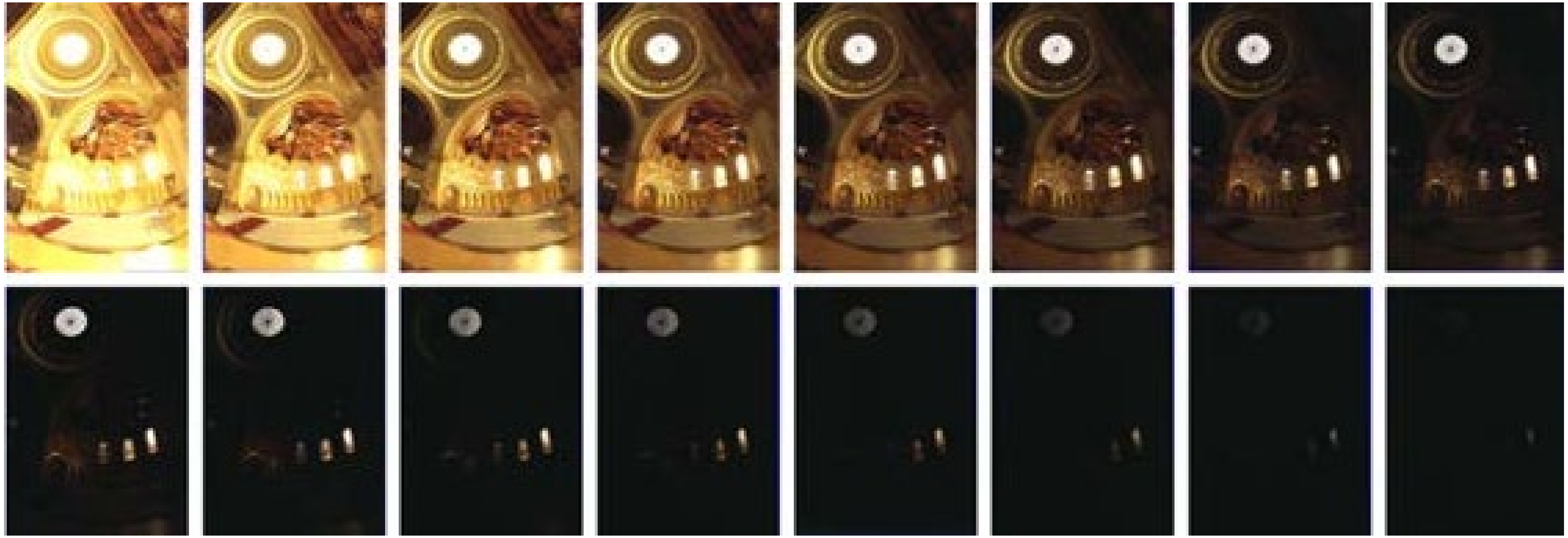
→ Combine them!

Sources: [CMU 15-463]


HYATT

HDRI: One of the first (and most used) methods

1. Exposure bracketing: Capture multiple LDR images at different exposures



This is called a LDR exposure stack

2. Merging: Combine them into a single HDR image

Sources: [CMU 15-463]

HDRI: Multi-bracketing

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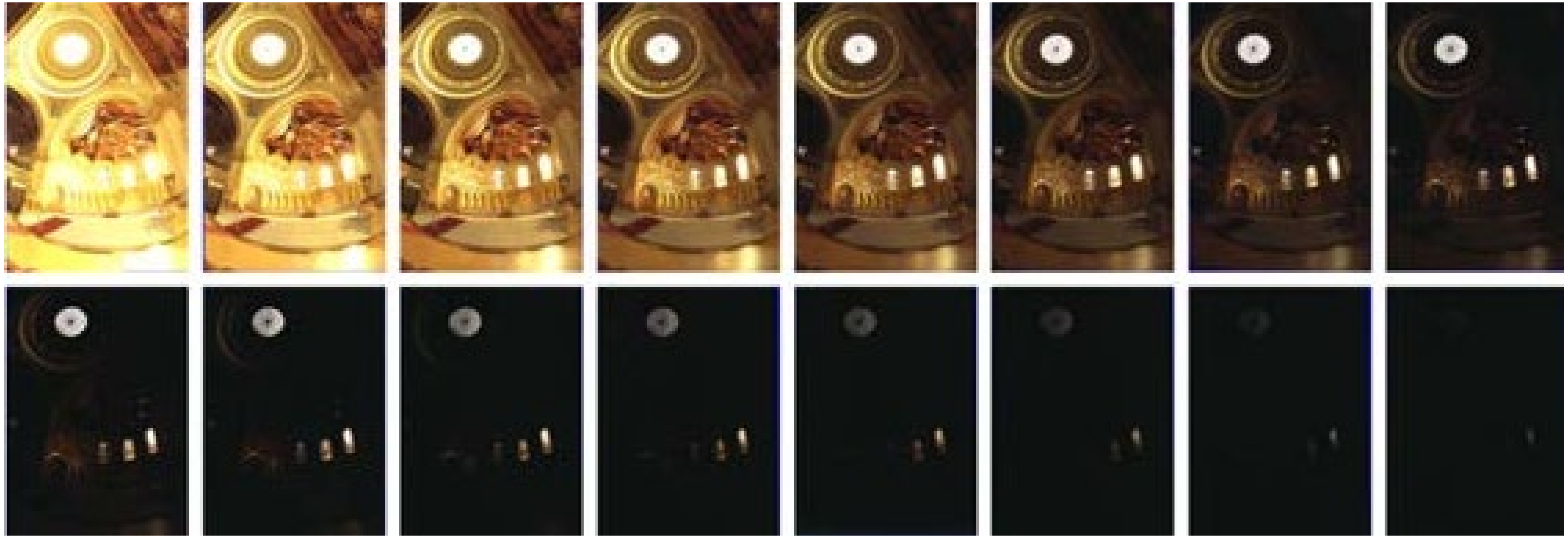

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Sources: CMU 15-463

Multi-bracketing: Ways to vary exposure

What is exposure?

Roughly speaking, the “brightness” of a captured image given a fixed scene. In other words, the amount of light reaching the image sensor (or film) during a given time period.

$$\text{Exposure} = \text{Gain} * \text{Irradiance} * \text{Time}$$

- **Irradiance** (into the sensor) is indirectly controlled by the aperture. Represents the amount of light that gets into the sensor.
- **Time** is controlled by the shutter speed.
- **Gain** is controlled by the ISO.

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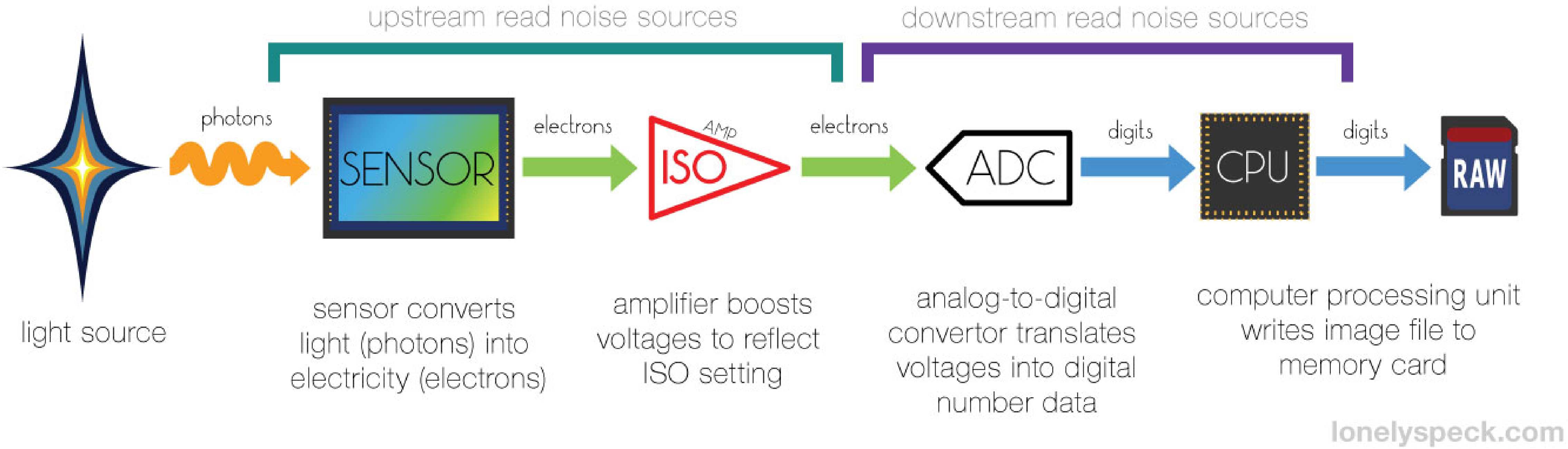
A brief note on ISO...

Sources: CMU 15-463

Side-effects of increasing ISO



Sources: CMU 15-463

Side-effects of increasing ISO

Image may become very grainy because noise is amplified.

ISO 80 ISO 800 ISO 1600

Multi-bracketing: Ways to vary exposure

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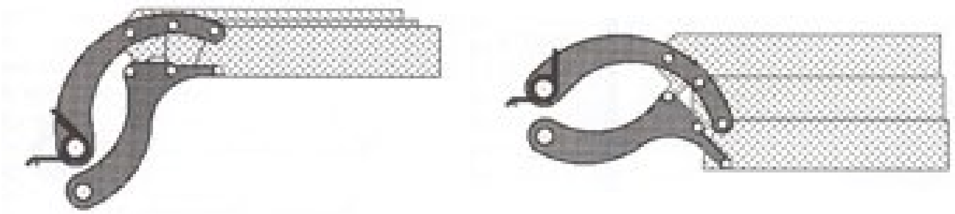
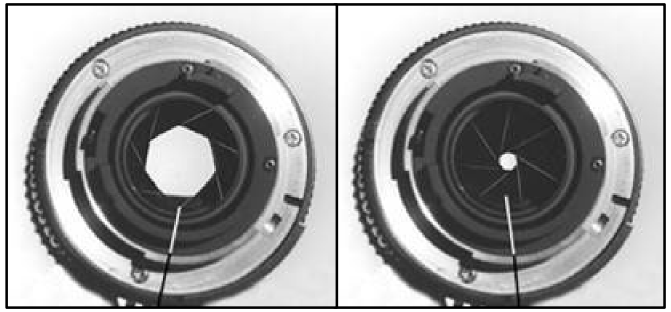
Multi-bracketing: Ways to vary exposure

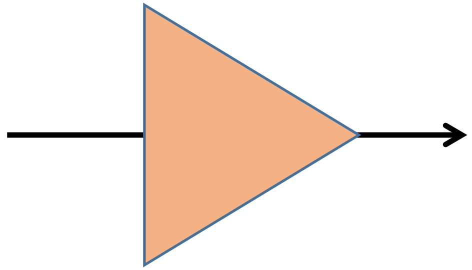
1. Shutter speed



2. F-stop (aperture, iris)



3. ISO



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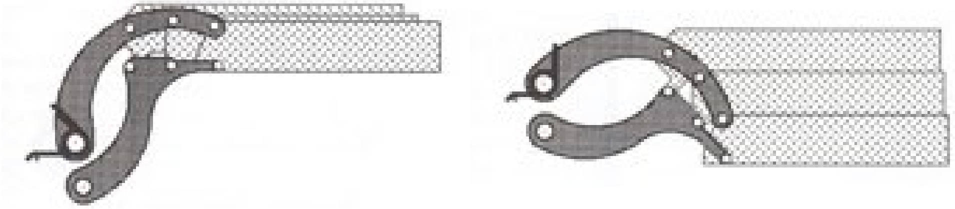

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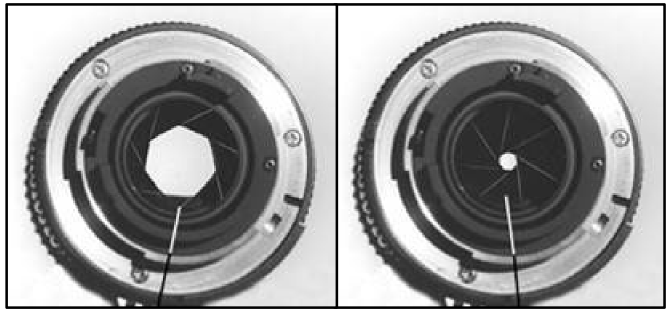

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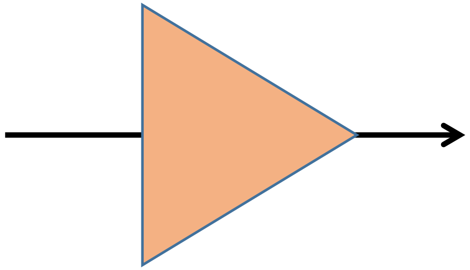
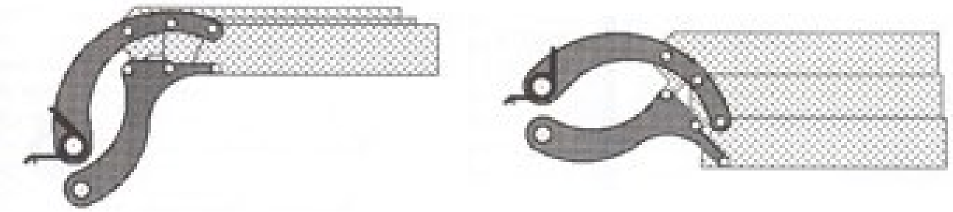


4. Neutral density (ND) filters

Sources: [CMU 15-463]

Multi-bracketing: Ways to vary exposure



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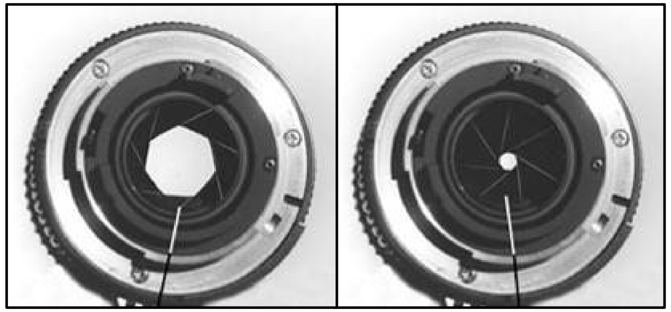
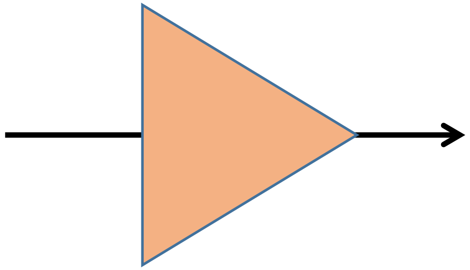


2. F-stop (aperture, iris)

Pros and cons of each?



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Sources: [CMU 15-463]

Multi-bracketing: Ways to vary exposure

1. Shutter speed
   - Range: about 30 sec to 1/4000 sec (6 orders of magnitude)
   - Pros: repeatable, linear
   - Cons: motion blur for long exposure
2. F-stop (aperture, iris)
   - Range: about f/0.98 to f/22 (3 orders of magnitude)
   - Pros: fully optical, no noise
   - Cons: changes depth of field
3. ISO
   - Range: about 100 to 1600 (1.5 orders of magnitude)
   - Pros: no movement at all
   - Cons: noise
4. Neutral density (ND) filters
   - Range: up to 6 densities (6 orders of magnitude)
   - Pros: works with strobe/flash
   - Cons: not perfectly neutral (color shift), extra glass (interreflections, aberrations), need to touch camera (shake)

Sources: CMU 15-463

HDRI: One of the first (and most used) methods

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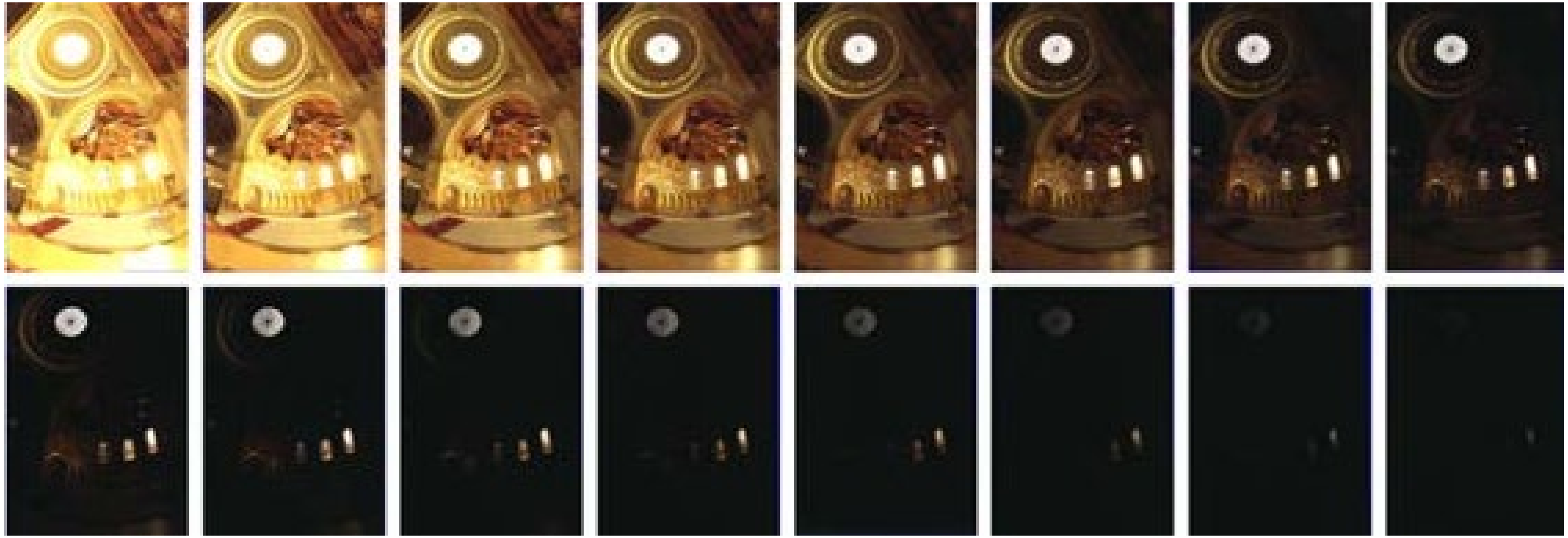

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Sources: CMU 15-463

Multi-bracketing: Exposure bracketing with shutter speed

Note: Shutter times usually obey a power series – each "stop" is a factor of 2

1/4, 1/8, 1/15, 1/30, 1/60, 1/125, 1/250, 1/500, 1/1000 sec

usually really is

1/4, 1/8, 1/16, 1/32, 1/64, 1/128, 1/256, 1/512, 1/1024 sec

Questions:

1. How many exposures?
2. What exposures?

Multi-bracketing: Exposure bracketing with shutter speed

Note: shutter times usually obey a power series – each "stop" is a factor of 2

1/4, 1/8, 1/15, 1/30, 1/60, 1/125, 1/250, 1/500, 1/1000 sec

usually really is

1/4, 1/8, 1/16, 1/32, 1/64, 1/128, 1/256, 1/512, 1/1024 sec

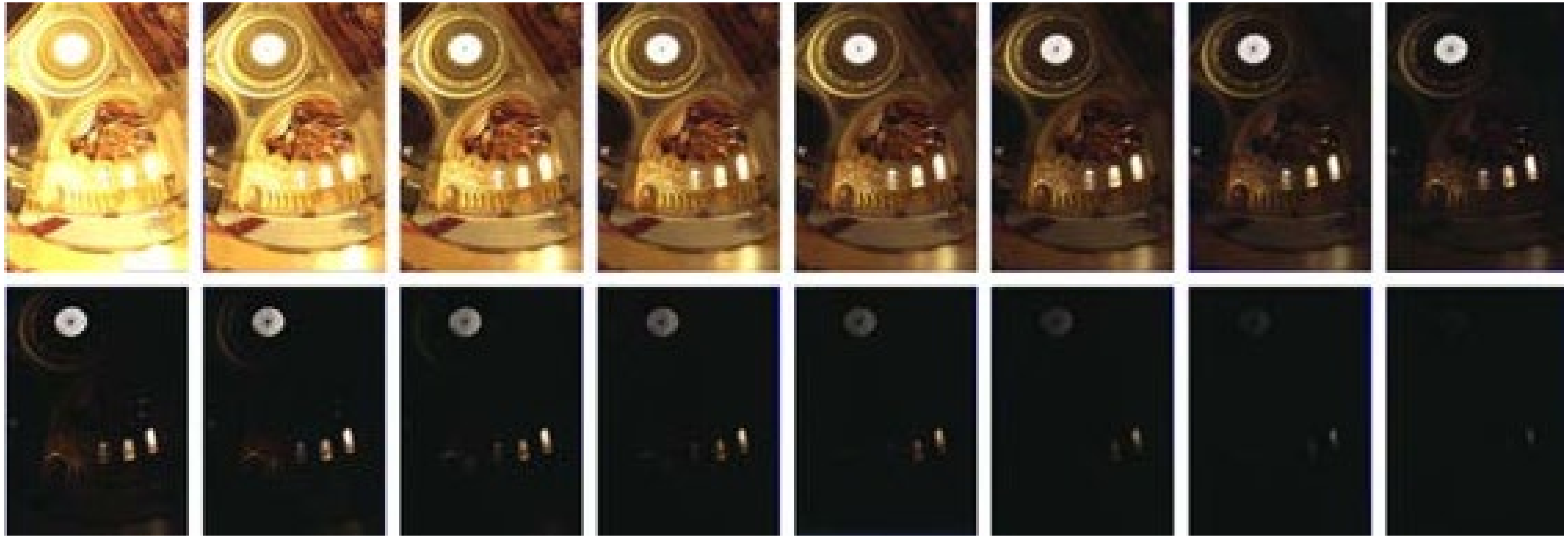
Questions:

1. How many exposures?
2. What exposures?

Answer: Depends on the scene, but a good default is 5 exposures, the metered exposure and +/- 2 stops around that.

HDRI: Multi-bracketing

1. Exposure bracketing: Capture multiple LDR images at different exposures



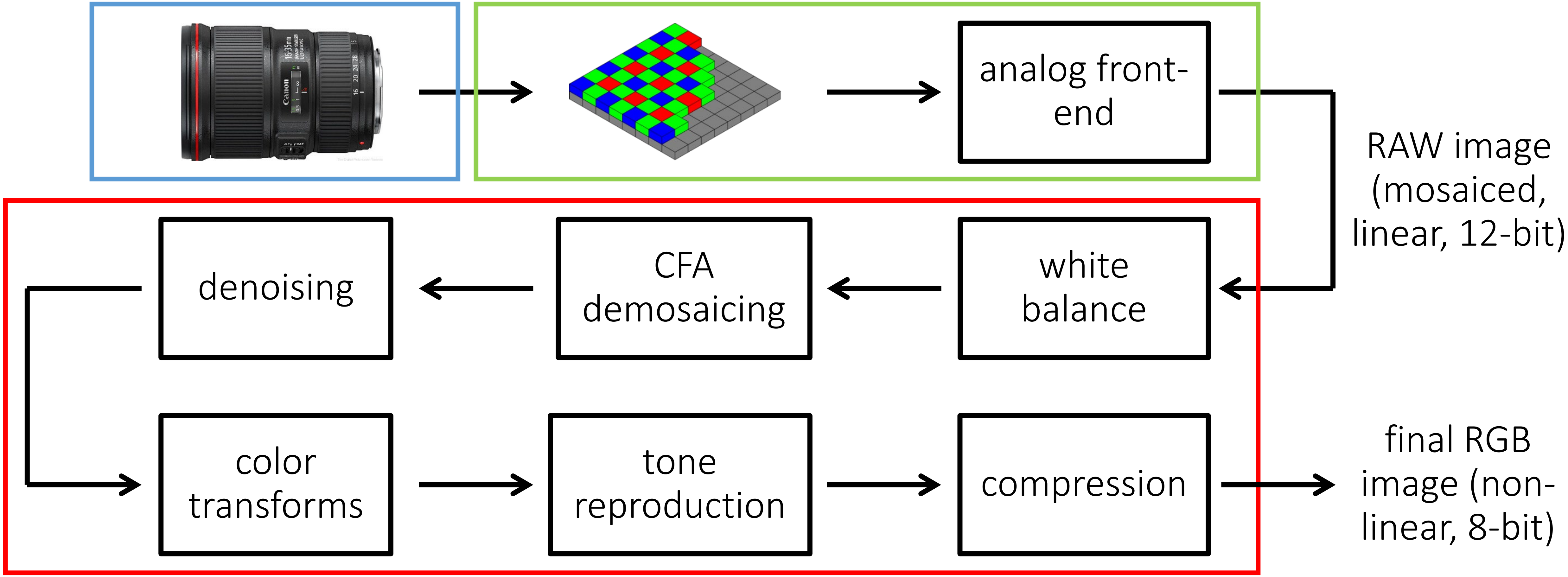
2. Merging: Combine them into a single HDR image

Sources: [CMU 15-463]

Multi-bracketing: Merging

Remember the image processing pipeline...

The sequence of image processing operations applied by the camera's image signal processor (ISP) to convert a RAW image into a "conventional" image.



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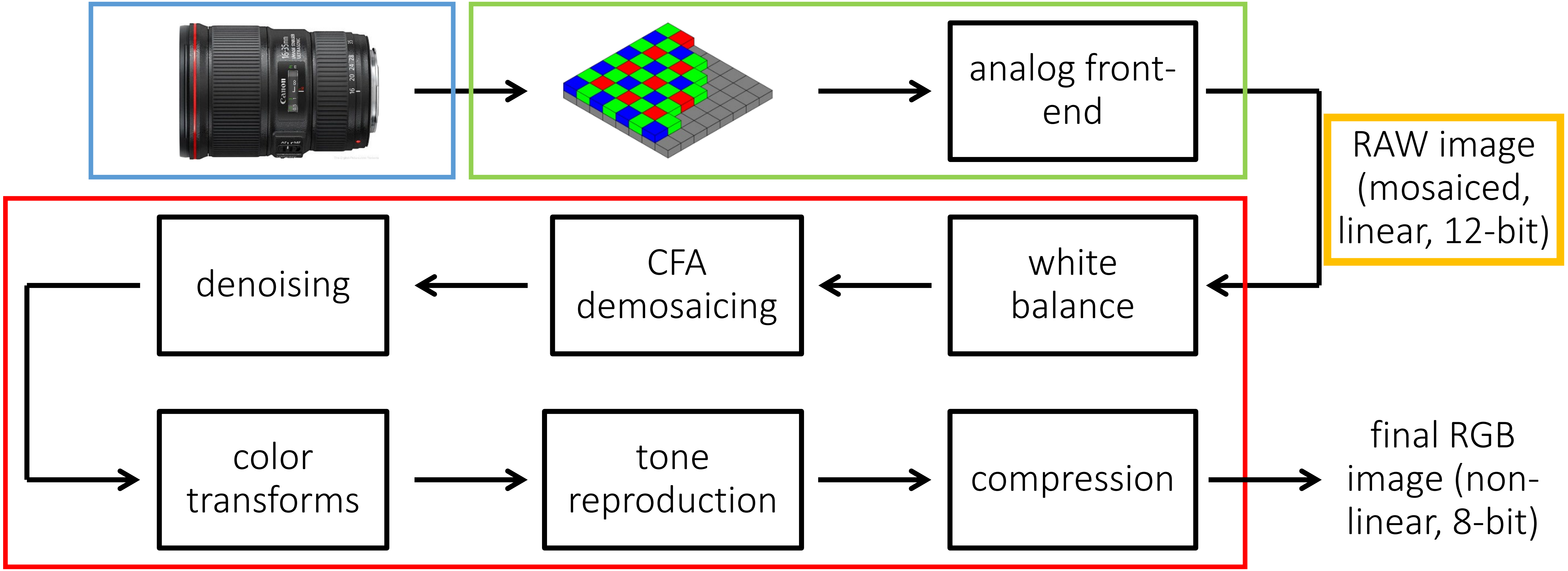
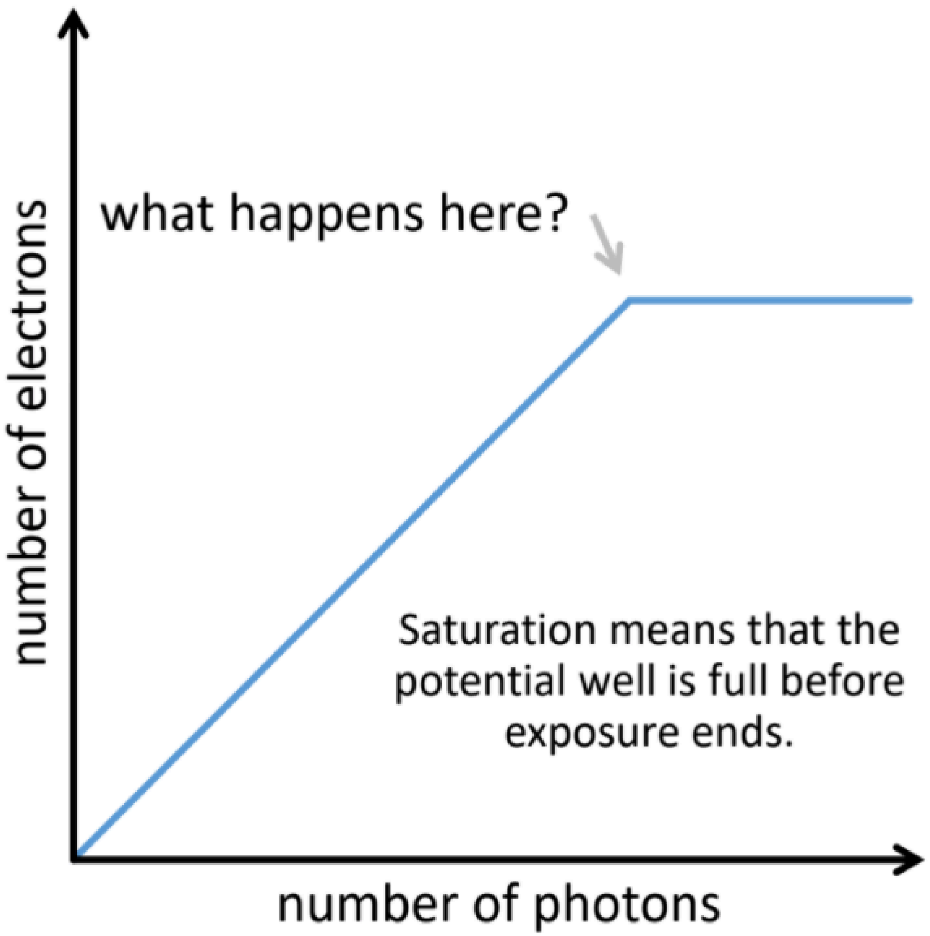
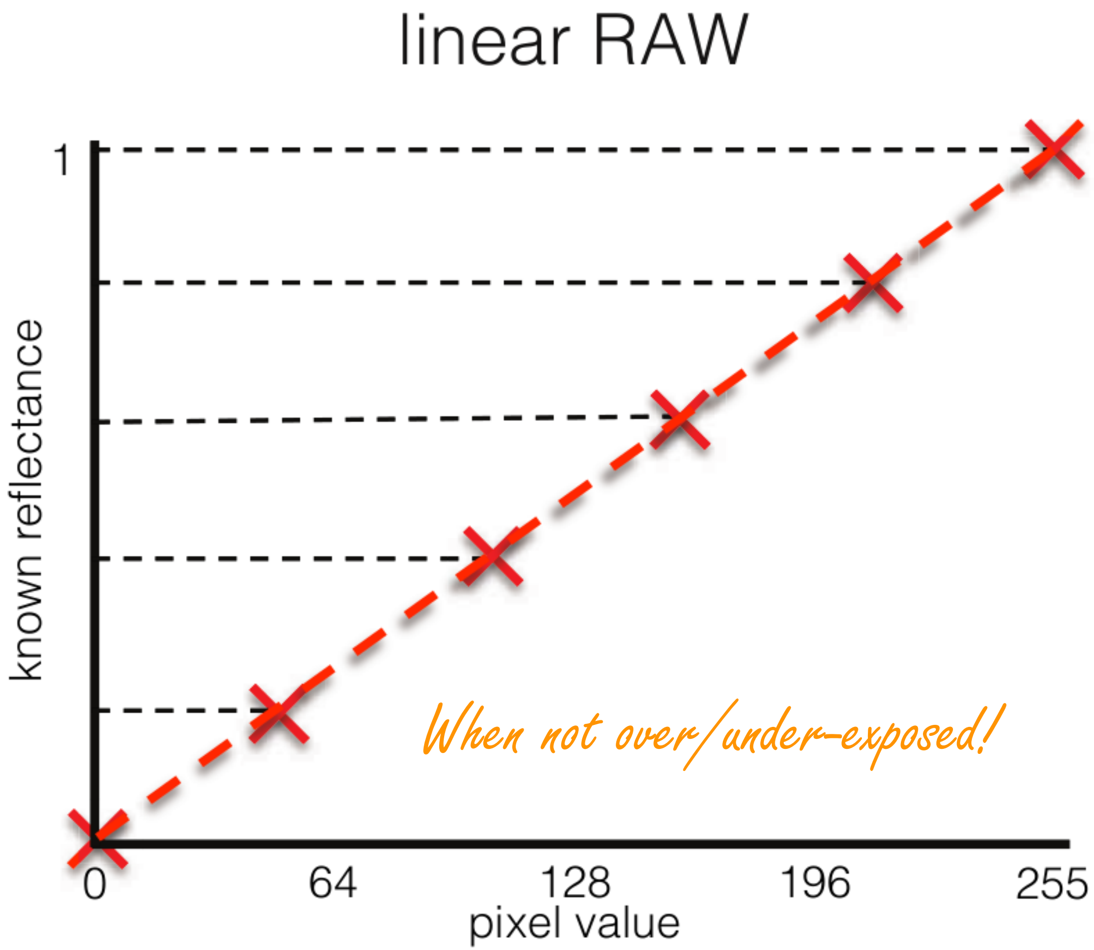


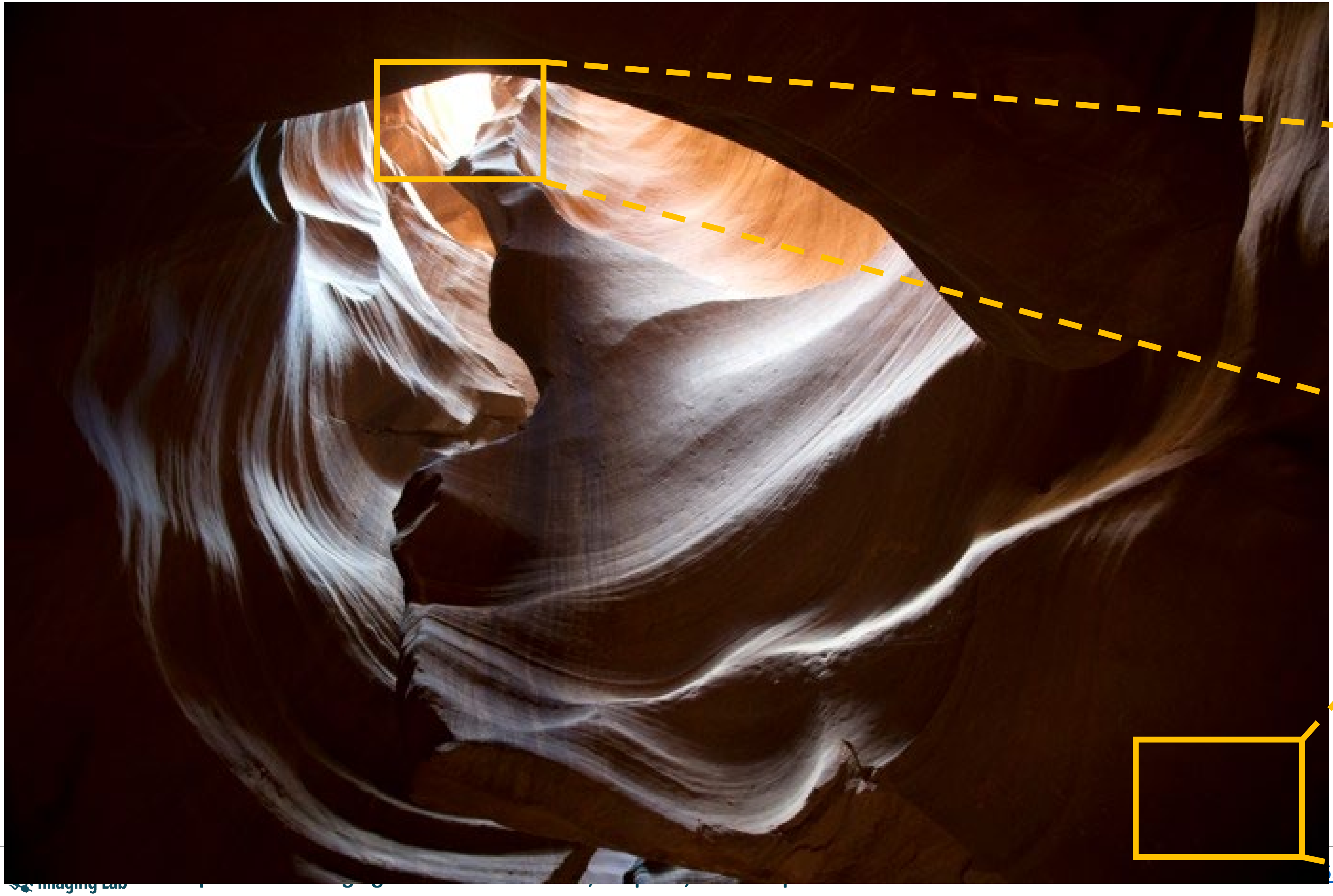
(reminder) RAW images have a linear response curve





Sources: [CMU 15-463]

Under- and over-exposure: An example



in highlights we are limited by clipping

in shadows we are limited by noise

RAW (linear) image formation model

What we seek to obtain

Real scene radiance for image pixel (x,y): $L(x,y)$

Exposure time:

t_5

t_4

t_3

t_2

t_1

What is an expression for the image $I_{linear}(x,y)$ as a function of $L(x,y)$?

Sources: [CMU 15-463](CMU 15-463)

RAW (linear) image formation model

Real scene radiance for image pixel (x,y): $L(x,y)$

Exposure time:

t_5 t_4 t_3 t_2 t_1

What is an expression for the image $I_{linear}(x,y)$ as a function of $L(x,y)$?

$$I_{linear}(x,y) = \text{clip}[\, t_i \cdot L(x,y) + \text{noise} \,]$$

How can we merge these images into one?

Sources: [CMU 15-463]

Multi-bracketing: Merging RAW (linear) images

For each pixel:

1. Find “valid” images

2. Weight valid pixel values appropriately

3. Form a new pixel value as the weighted average of valid pixel values

What are “valid” images?

t_5 t_4 t_3 t_2 t_1

Multi-bracketing: Merging RAW (linear) images

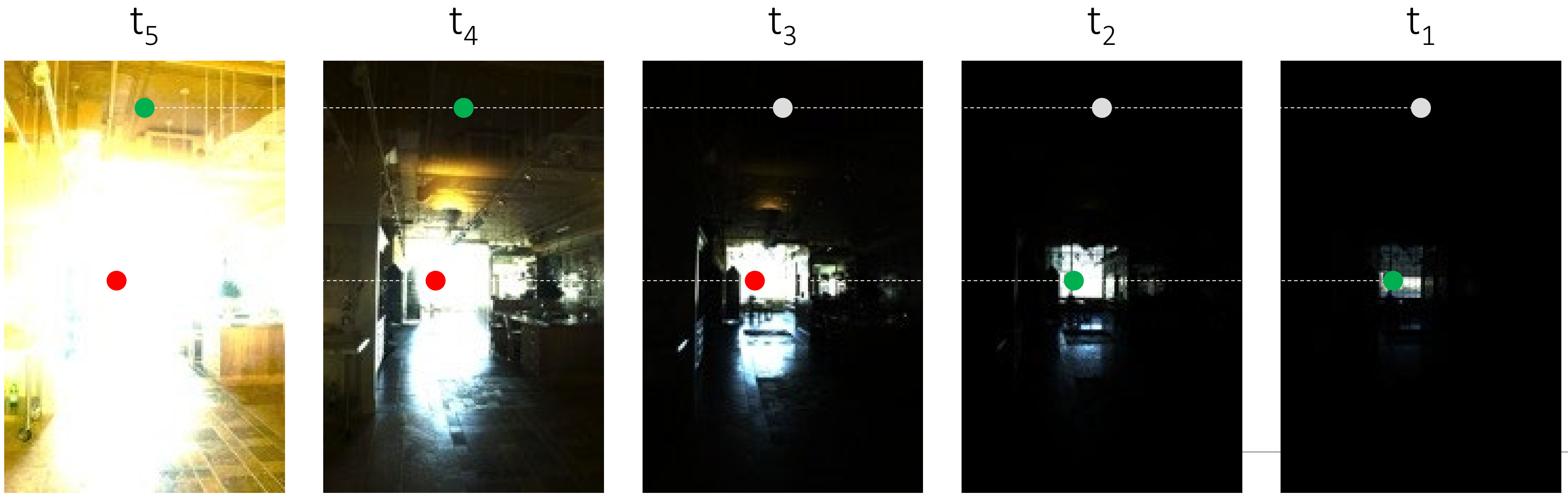
For each pixel:

1. Find “valid” images ⟵ (noise) 0.05 < pixel < 0.95 (clipping)
2. Weight valid pixel values appropriately
3. Form a new pixel value as the weighted average of valid pixel values

noise

valid

clipped



Multi-bracketing: Merging RAW (linear) images

For each pixel:

1. Find “valid” images ⟵ (noise) 0.05 < pixel < 0.95 (clipping)

2. Weight valid pixel values appropriately ⟵ (pixel value) / t_i

3. Form a new pixel value as the average of valid pixel values

t_5

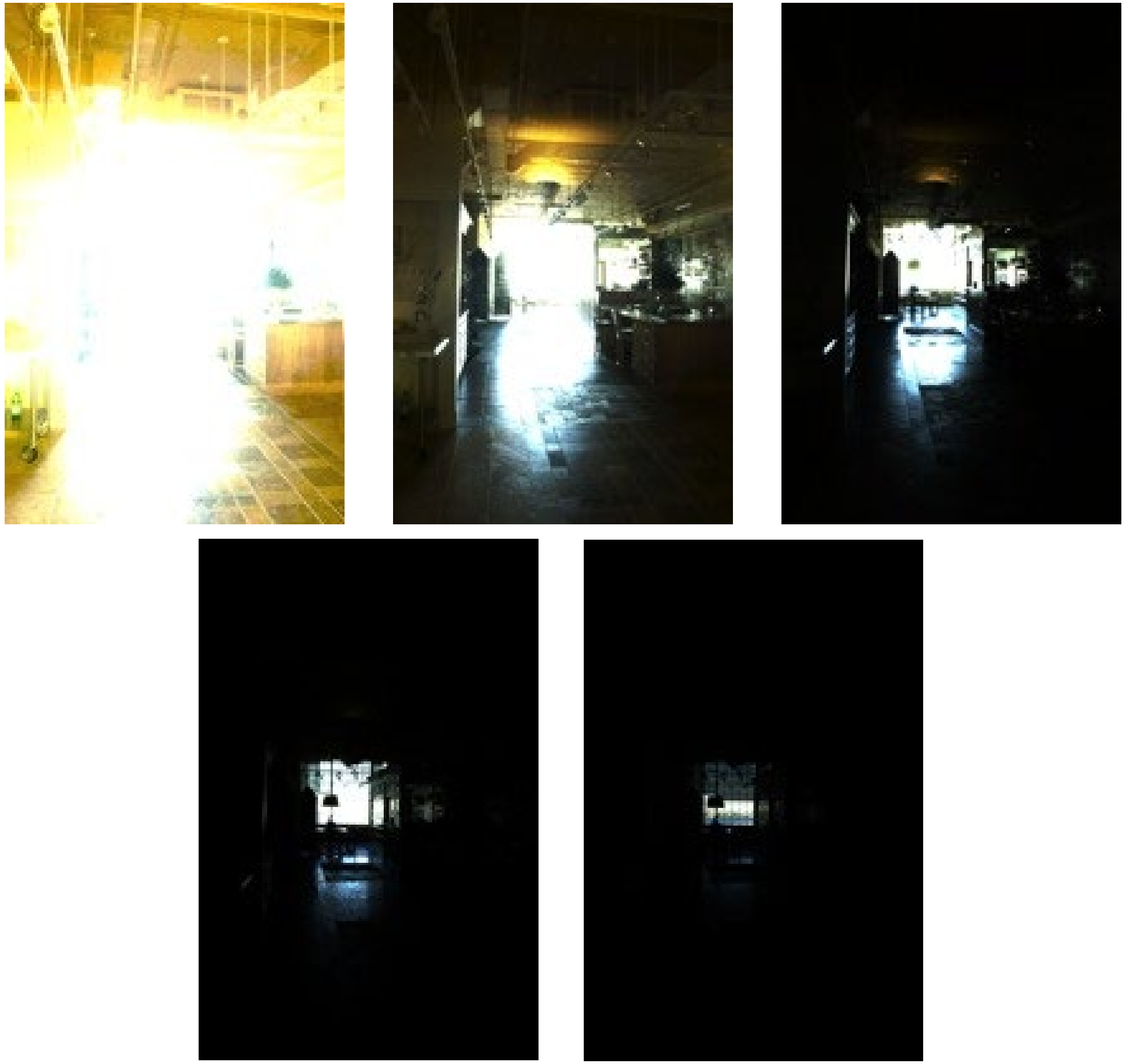
t_4

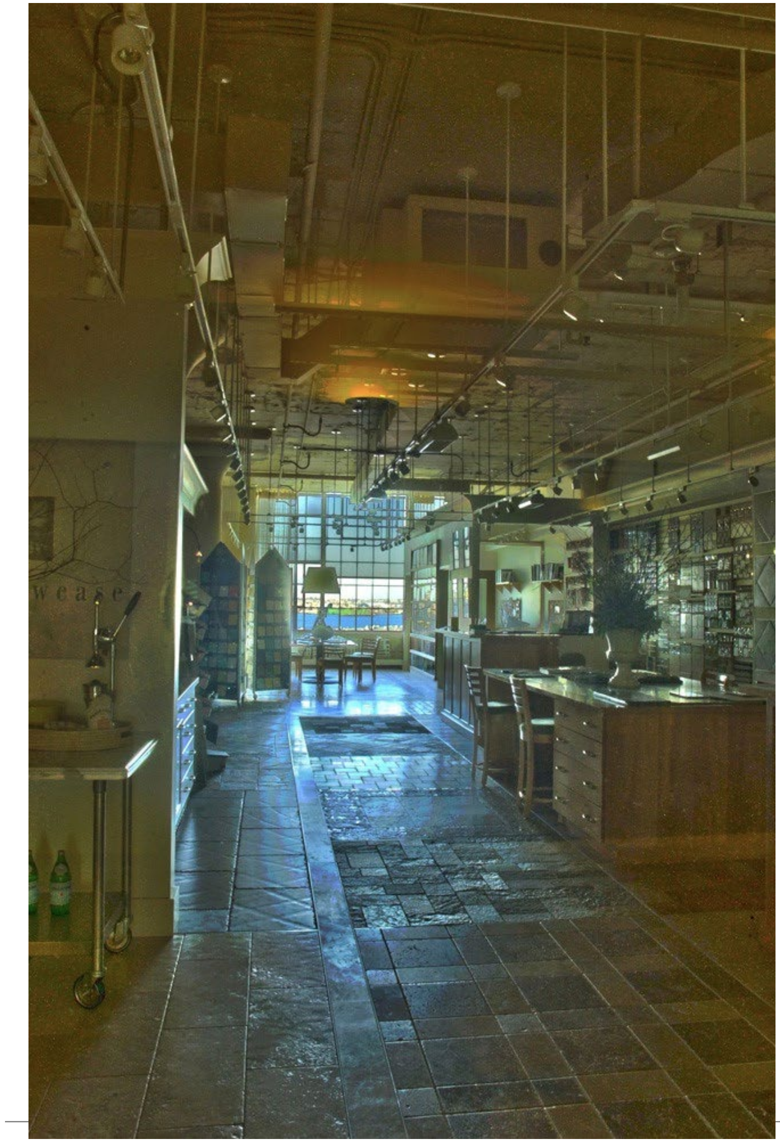
t_3

t_2

t_1

Multi-bracketing: Merging result (after tonemapping)





Sources: [CMU 15-463]

Multi-bracketing for HDRI

What do we do if we cannot use RAW?

Sources: CMU 15-463

Multi-bracketing for HDRI

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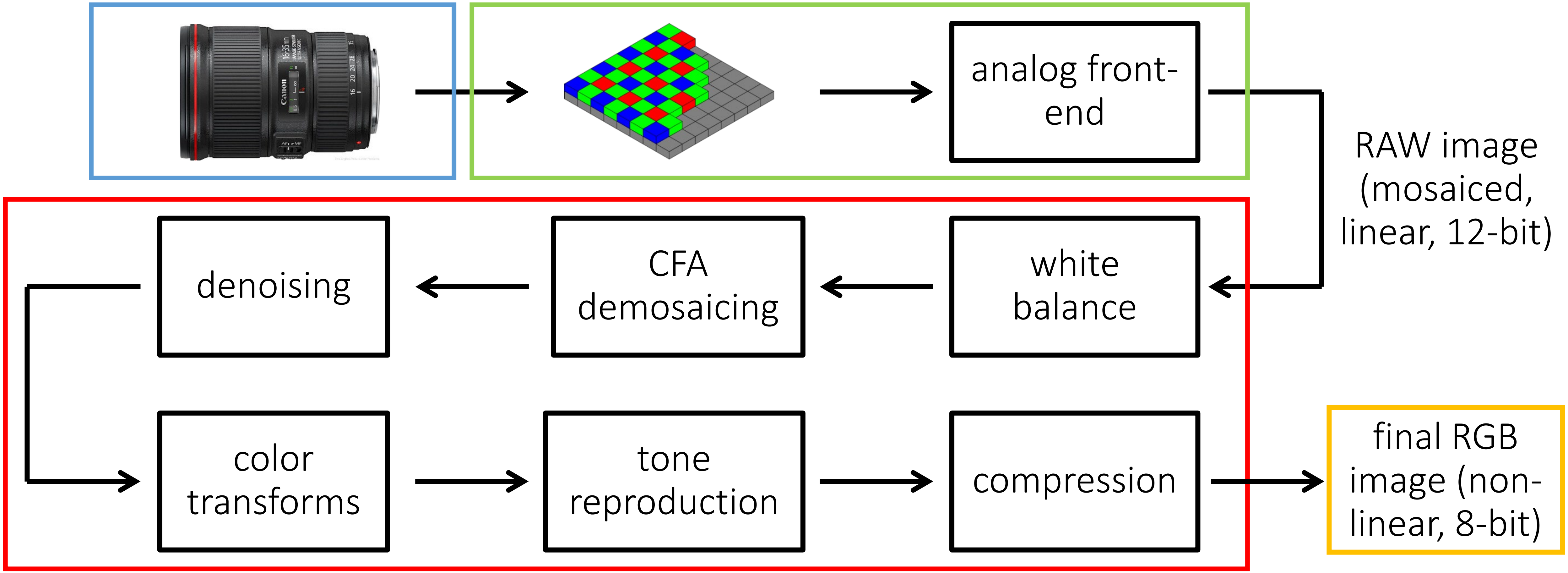


Sources: [CMU 15-463]

Multi-bracketing for HDRI

What do we do if we cannot use RAW?

Non-linear image formation model:

$$I_{linear}(x,y) = \text{clip}[\ t_i \cdot L(x,y) + \text{noise}\ ]$$

$$I_{non\text{-}linear}(x,y) = f[\ I_{linear}(x,y)\ ]$$

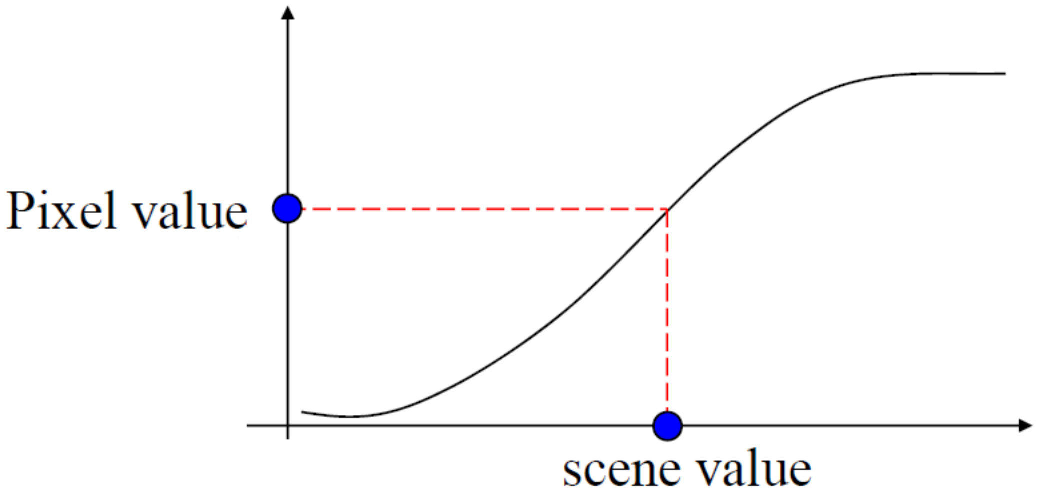
We need to perform **radiometric calibration**.

Sources: [CMU 15-463]

Radiometric calibration

It is the process of measuring the camera's response curve.

- Camera Response Curve
  - Relation between amount of incoming light and image pixel values of a digital camera



Radiometric calibration

It is the process of measuring the camera’s response curve.

Three ways to do radiometric calibration of a camera:

1. Take images of scenes with **different irradiance** while keeping exposure the same.

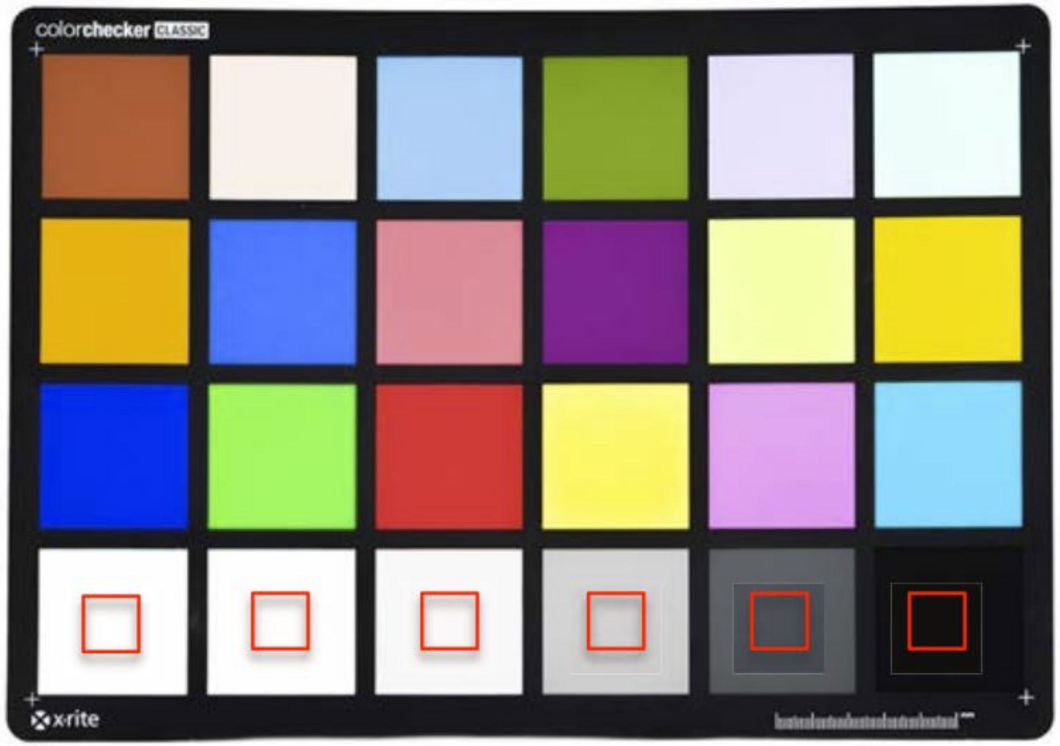
2. Takes images under **different exposures** while keeping irradiance the same.

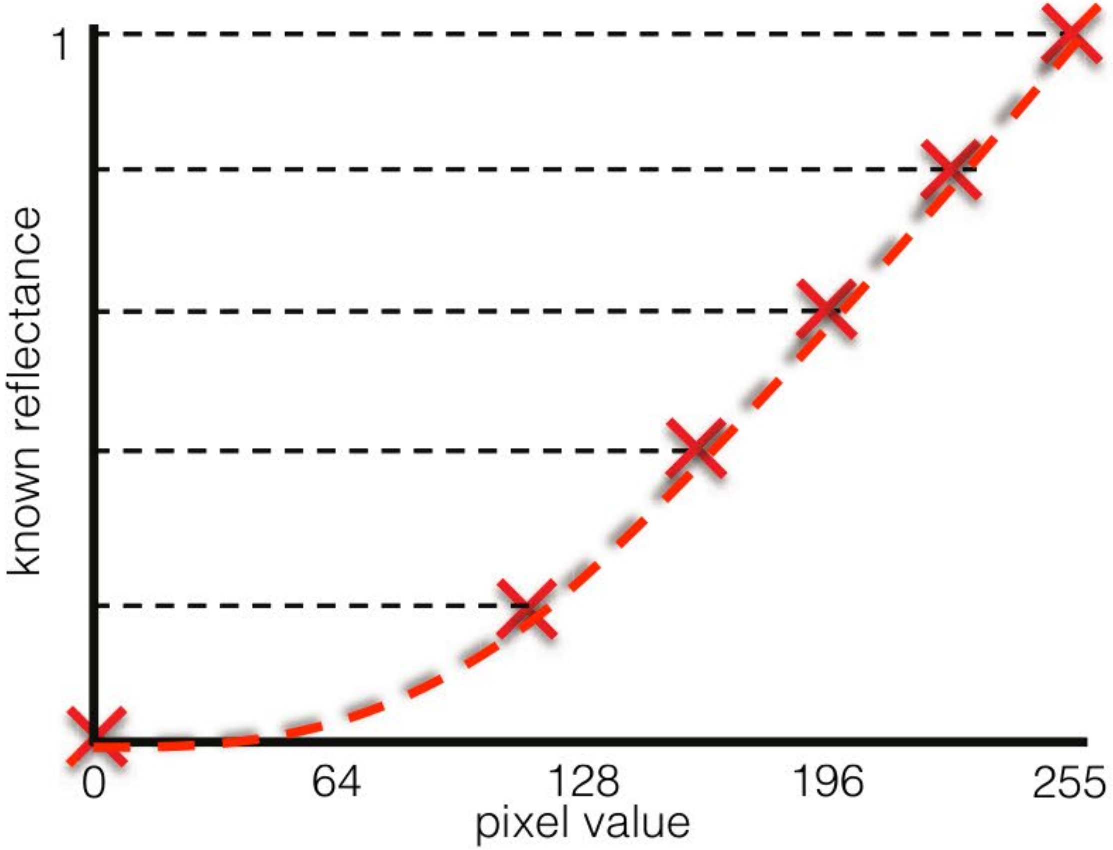
3. Takes images of scenes with **different irradiance and under different exposures**.

Sources: CMU 15-463

1. Same camera exposure, varying scene irradiance

Colorchecker: Tool for radiometric & color calibration.





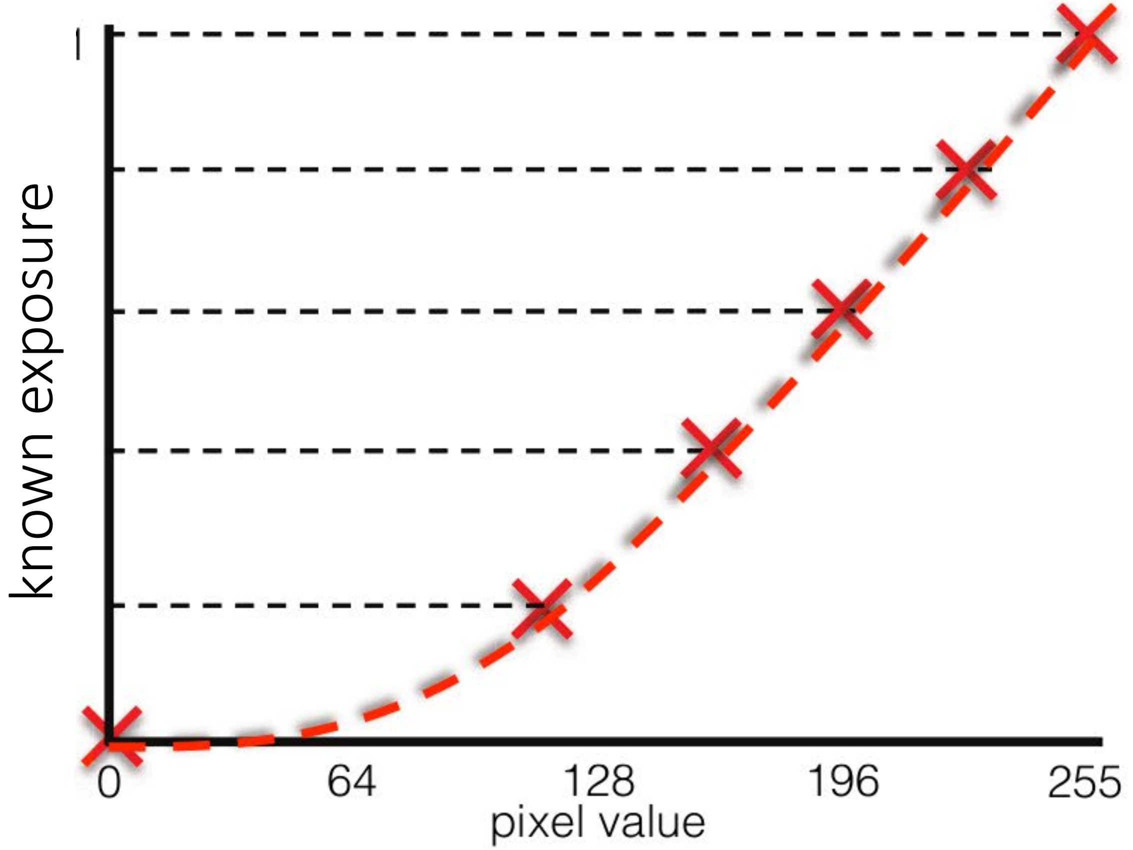
Patches at bottom row have reflectance that increases linearly.

Different values correspond to patches of increasing reflected irradiance.

Sources: [CMU 15-463]

2. Same scene irradiance, varying camera exposure

Colorchecker: Tool for radiometric & color calibration.



All points on (the white part of) the target have the same reflectance.

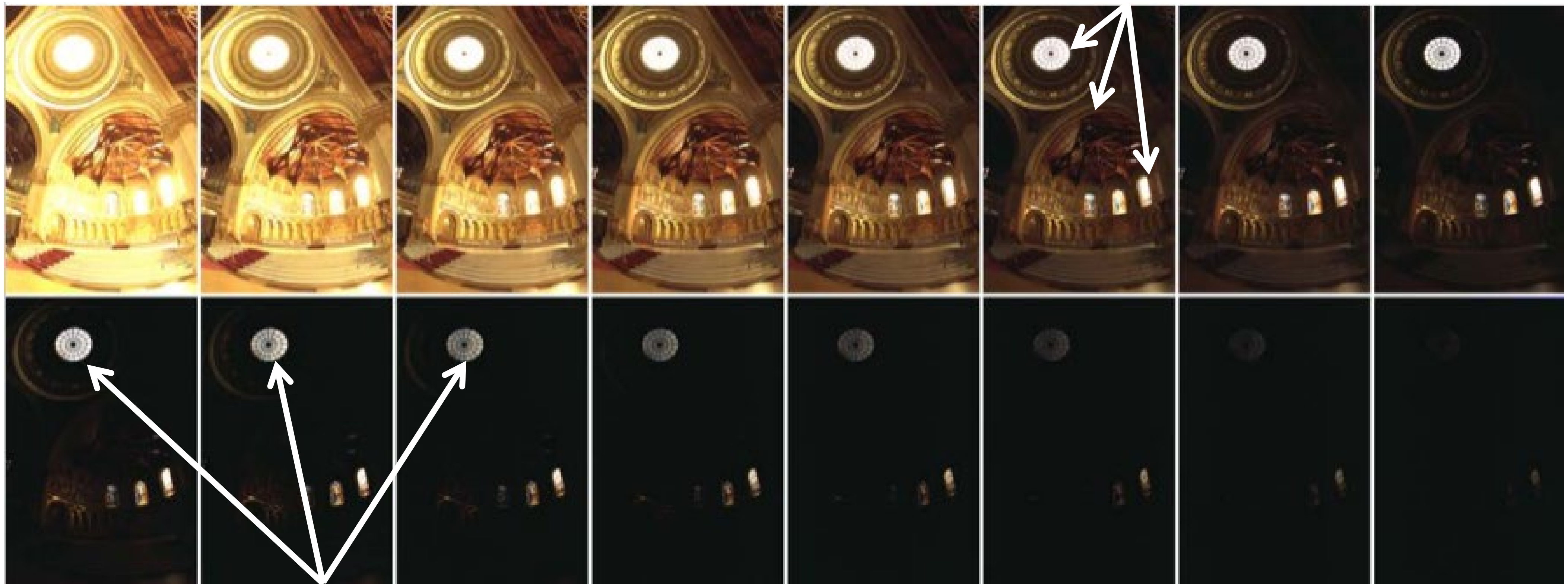
Different values correspond to images taken under increasing camera exposure.

Sources: [CMU 15-463]

3. Varying both scene irradiance and camera exposure

This can be done using the LDR exposure stack itself
→ [Debevec and Malik 1997] Recovering High Dynamic Range Radiance Maps from Photographs. SIGGRAPH.

Same camera exposure, different scene irradiance



Same scene irradiance, different camera exposure

Sources: CMU 15-463

Radiometric calibration

It is the process of measuring the camera's response curve.

Three ways to do radiometric calibration of a camera:

1. Take images of scenes with different irradiance while keeping exposure the same.
2. Takes images under different exposures while keeping irradiance the same.
3. Takes images of scenes with different irradiance and under different exposures.

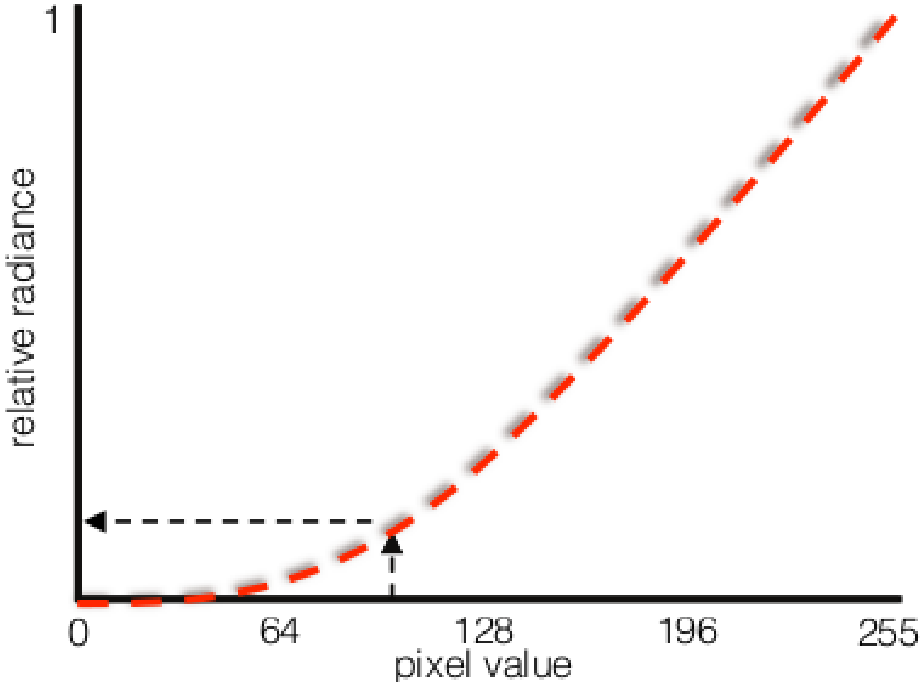
Once we have the camera's response curve (f), how do we proceed?

Sources: [CMU 15-463]

Non-linear image formation model

Real scene radiance for image pixel (x,y): L(x, y)

Exposure time: t_i



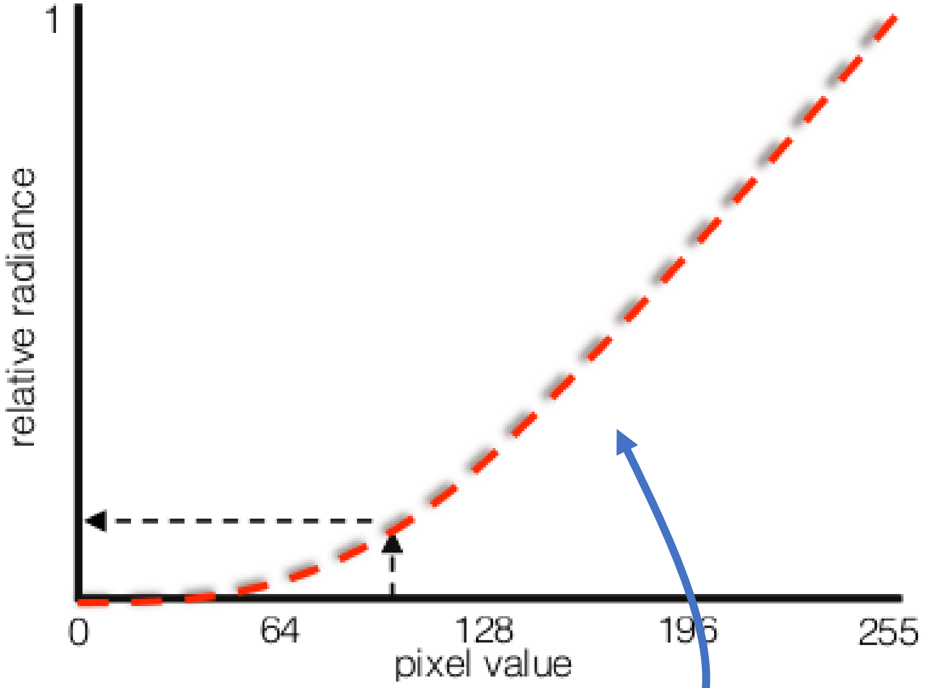
$$I_{linear}(x,y) = \text{clip}[\, t_i \cdot L(x,y) + \text{noise} \,]$$

$$I_{non\text{-}linear}(x,y) = f[\, I_{linear}(x,y) \,]$$

Non-linear image formation model

Real scene radiance for image pixel (x,y): $L(x, y)$

Exposure time: t_i



$$I_{linear}(x,y) = \text{clip}[\ t_i \cdot L(x,y) + \text{noise}\ ]$$

$$I_{non\text{-}linear}(x,y) = f[\ I_{linear}(x,y)\ ]$$

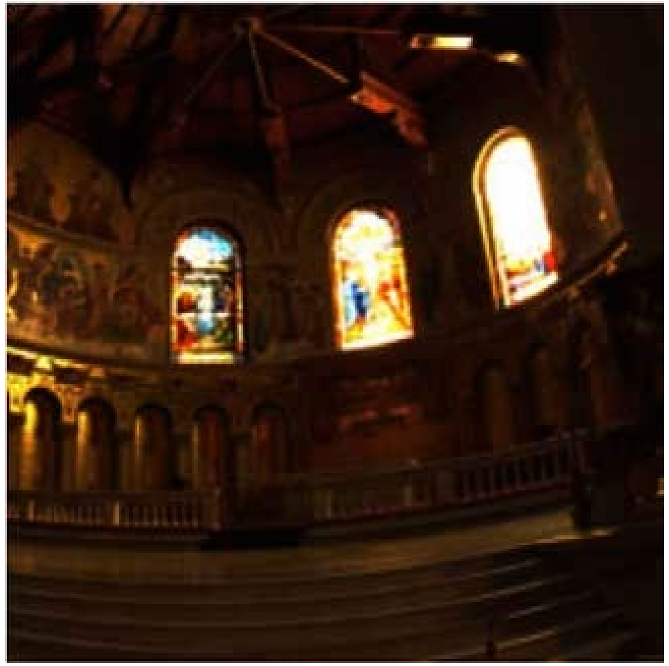
If we undo the non-linearity, we are back to I_{linear} as before... ⇨ f^{-1}

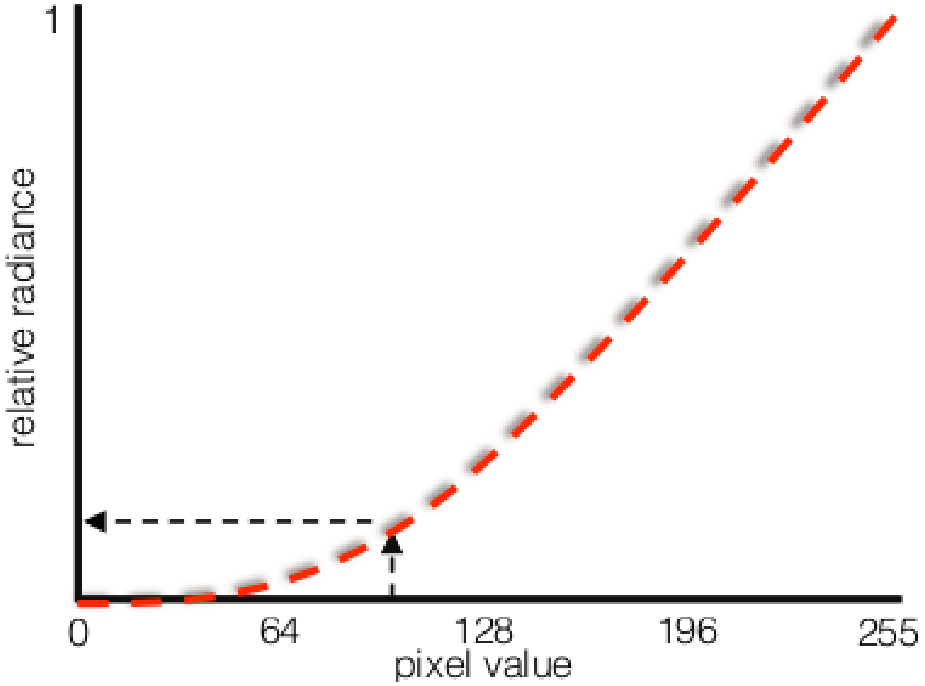
Sources: CMU 15-463

Multi-bracketing: Merging with non-linear images

Real scene radiance for image pixel (x,y): L(x, y)

Exposure time: t_i





$I_{linear}(x,y) = \text{clip}[\ t_i \cdot L(x,y) + \text{noise}\ ]$

$I_{non\text{-}linear}(x,y) = f[\ I_{linear}(x,y)\ ]$

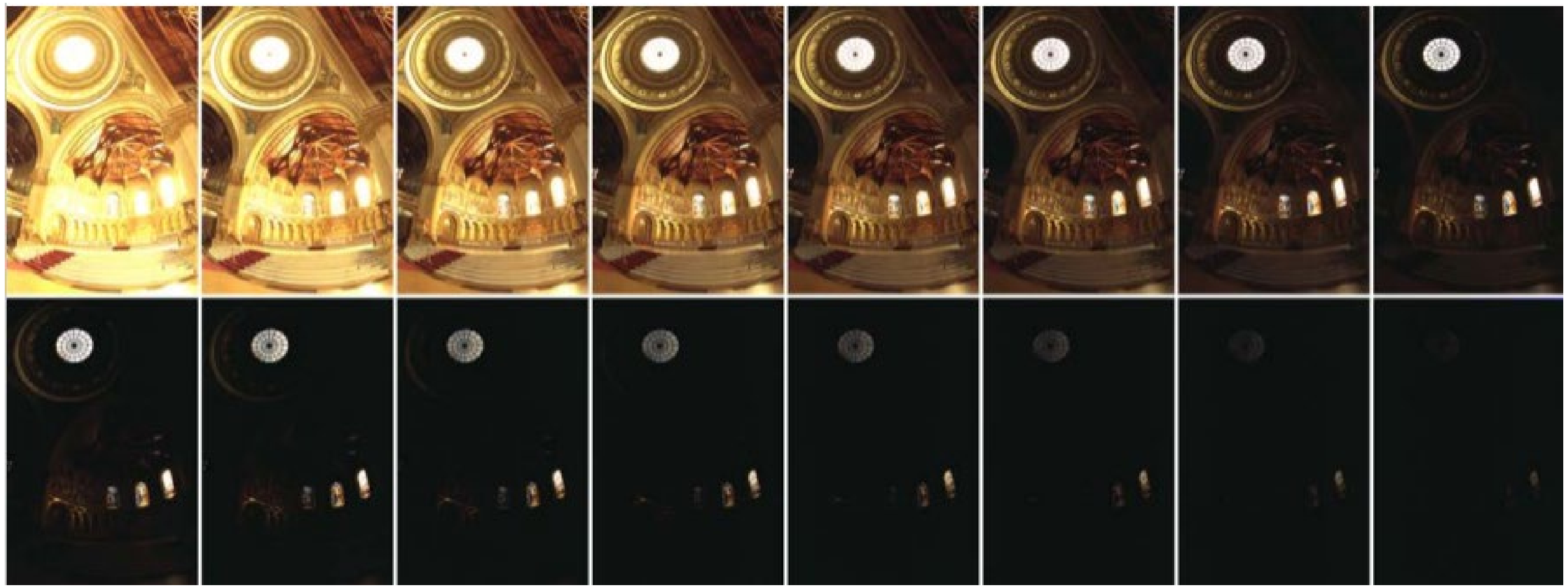
$I_{est}(x,y) = f^{-1}[\ I_{non\text{-}linear}(x,y)\ ]$

Sources: CMU 15-463

Multi-bracketing: Merging with non-linear images

Linearization:

$$I_{\text{non-linear}}(x,y) = f[\ I_{\text{linear}}(x,y)\ ]$$



$$I_{\text{est}}(x,y) = f^{-1}[\ I_{\text{non-linear}}(x,y)\ ]$$

Multi-bracketing: Merging with non-linear images

1. Calibrate response curve
2. Linearize images

For each pixel:

3. Find “valid” images ⟵ (noise) $0.05 < \text{pixel} < 0.95$ (clipping)
4. Weight valid pixel values appropriately ⟵ (pixel value) / t_i
5. Form a new pixel value as the weighted average of valid pixel values

Same steps as in the RAW case.

Sources: [CMU 15-463]

Multi-bracketing for HDRI

What do we do if we cannot use RAW?

We need to perform **radiometric calibration.**

A note on radiometric calibration and camera response...

Sources: [CMU 15-463]

Relative vs absolute radiance

The final HDR image gives radiance only up to a global scale → Relative radiance

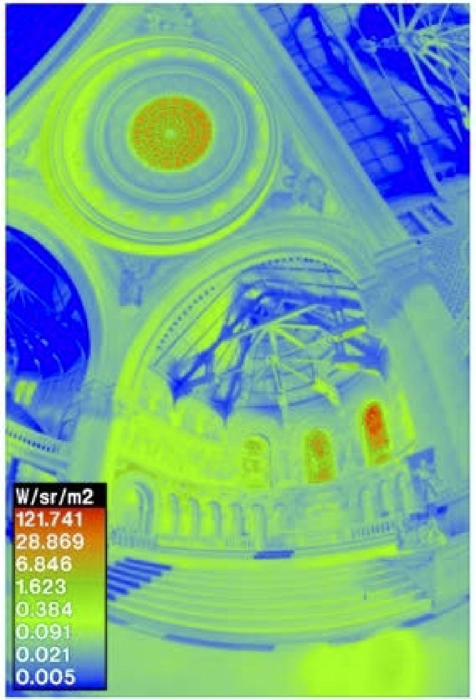
- If we know the exact radiance at a point, we can convert relative HDR image to absolute radiance map

HDR image
(relative radiance)

→

spotmeter (absolute radiance at one point)

→



absolute
radiance map

Sources: CMU 15-463

Relative vs absolute radiance

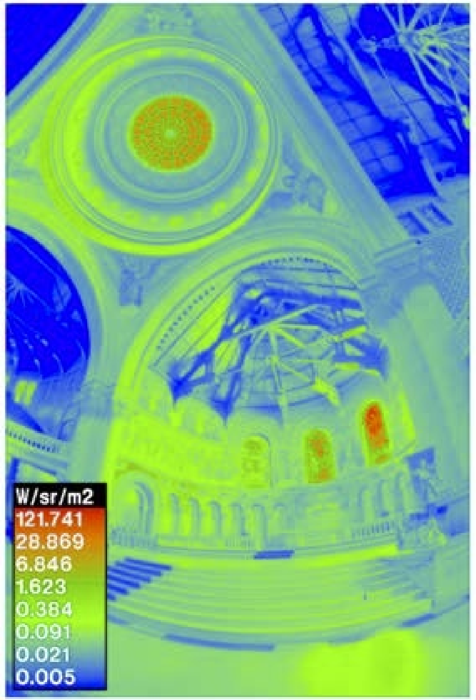
The final HDR image gives radiance only up to a global scale → Relative radiance
- If we know the exact radiance at a point, we can convert relative HDR image to absolute radiance map

HDR image
(relative radiance)

→

spotmeter (absolute radiance at one point)

→



absolute radiance map

* Note: It is not uncommon to represent HDR images in false color, given their large dynamic range.

Sources: CMU 15-463

Multi-bracketing for HDRI

What do we do if we cannot use RAW?

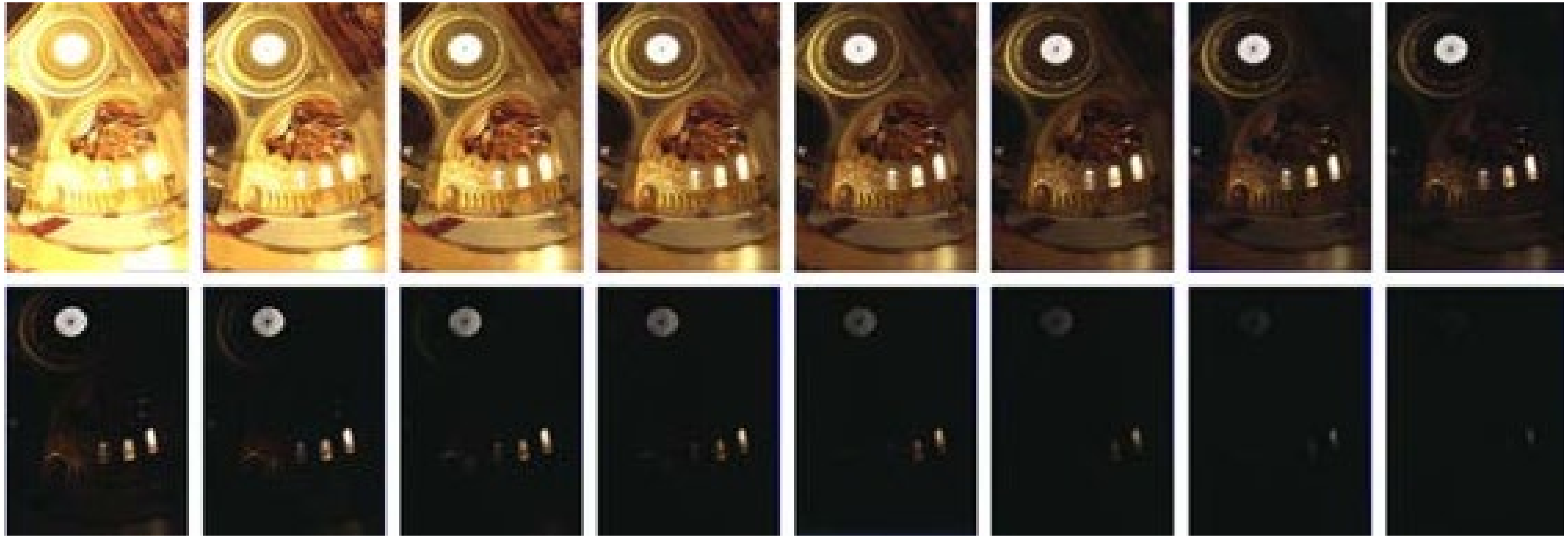
We need to perform **radiometric calibration**.

What do we do if we cannot use RAW nor perform radiometric calibration (we cannot measure the camera's response curve)?

We can try to find an approximate response curve.

HDRI: Multi-bracketing

1. Exposure bracketing: Capture multiple LDR images at different exposures



2. Merging: Combine them into a single HDR image

Sources: [CMU 15-463]

HDRI: Multi-bracketing

1. Exposure bracketing: Capture multiple LDR images at different exposures

Can you see potential problems?

2. Merging: Combine them into a single HDR image

Sources: [CMU 15-463]

HDRI: Multi-bracketing

1. Exposure bracketing: Capture multiple LDR images at different exposures

Can you see potential problems?

Motion!

Both scene and camera must be completely static.

If they are not -> Compute alignment as a postpro op.

2. Merging: Combine them into a single HDR image

HDRI: Is multi-bracketing the only option?

- Of course not (it is still the most common, though)

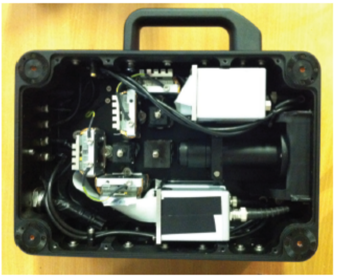
** Newer approaches try to address the problem of having a moving scene, try to do HDR in a single shot...*

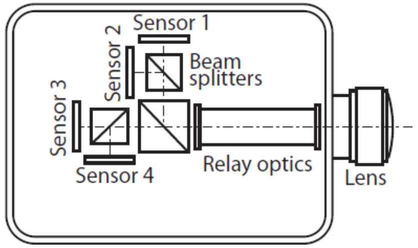
Sources: [CMU 15-463]

HDRI: Is multi-bracketing the only option?

- Of course not (it is still the most common, though)

- Hardware modifications to simultaneously capture different exposures of the same scene





[Kronander et al. 2013; Tocci et al. 2011; Batz et al. 2014; Schedl et al. 2014]

- Software approaches to merge images despite movement between them

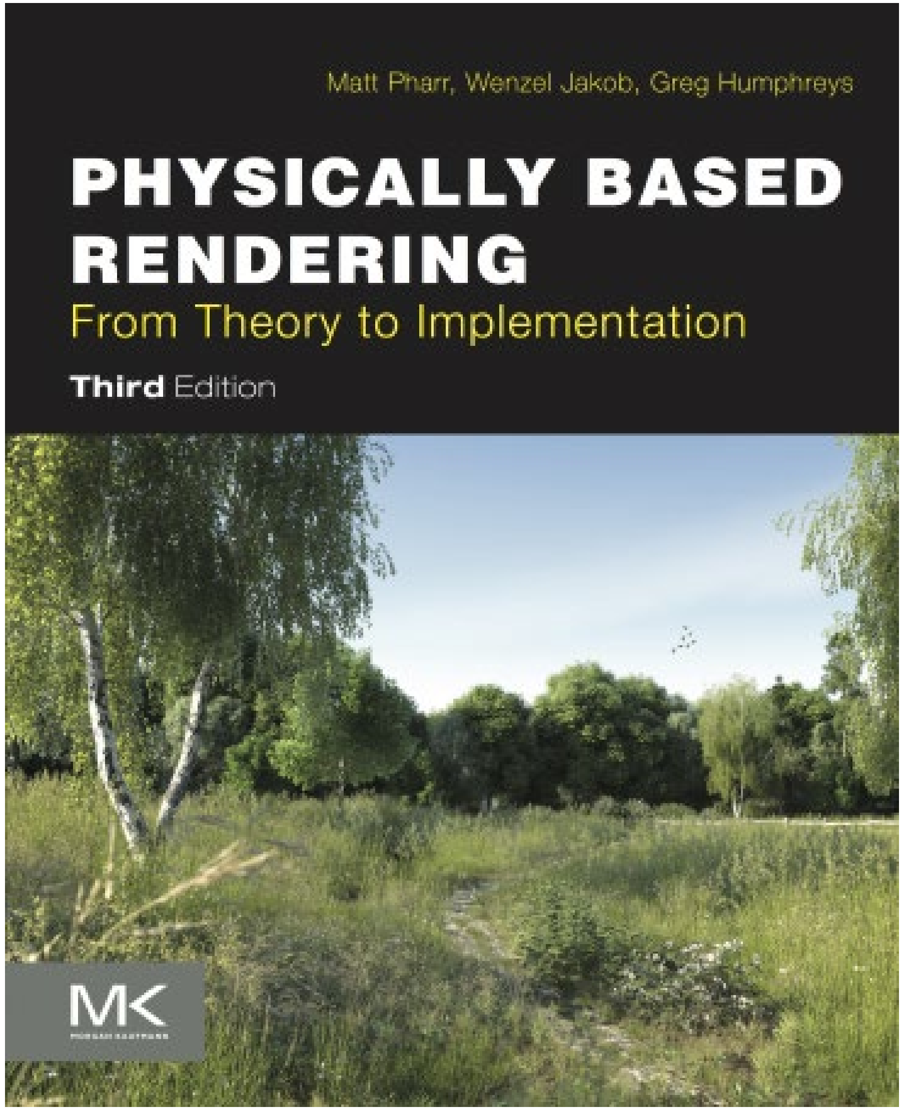
[Kalantari et al. 2013; Sen et al. 2012; Zimmer et al. 2011; Granados et al. 2013]

- Hybrid approaches (e.g., based on compressive sensing) & DL-based approaches

HDR images can also be rendered (instead of captured)

Physics-based renderers simulate radiance maps (relative or absolute)

- Their outputs are very often HDR images



Sources: [CMU 15-463]

HDR images: How are they stored?

- Most standard image formats store integer 8-bit images (e.g., JPEG)
- Some image formats store integer 12-bit or 16-bit images (e.g., TIFF)
- HDR images are floating point 32-bit or 64-bit images

- There are different HDR image formats:
  - Portable float map (.pfm)
  - Radiance format (.hdr)
  - OpenEXR format (.exr)

Sources: [CMU 15-463]

Alternative methods for “HDR imaging”

Sources: [CMU 15-463]

Exposure fusion

- Blend multiple exposures of the same scene into a single image
- Choose weights of the blend according to the *goodness* of each exposure
- **No HDR image is created during exposure fusion**
- Many phone cameras use this (or variants) for “HDR”: simplicity, speed, immediate appealing result

(a) Input images with corresponding weight maps

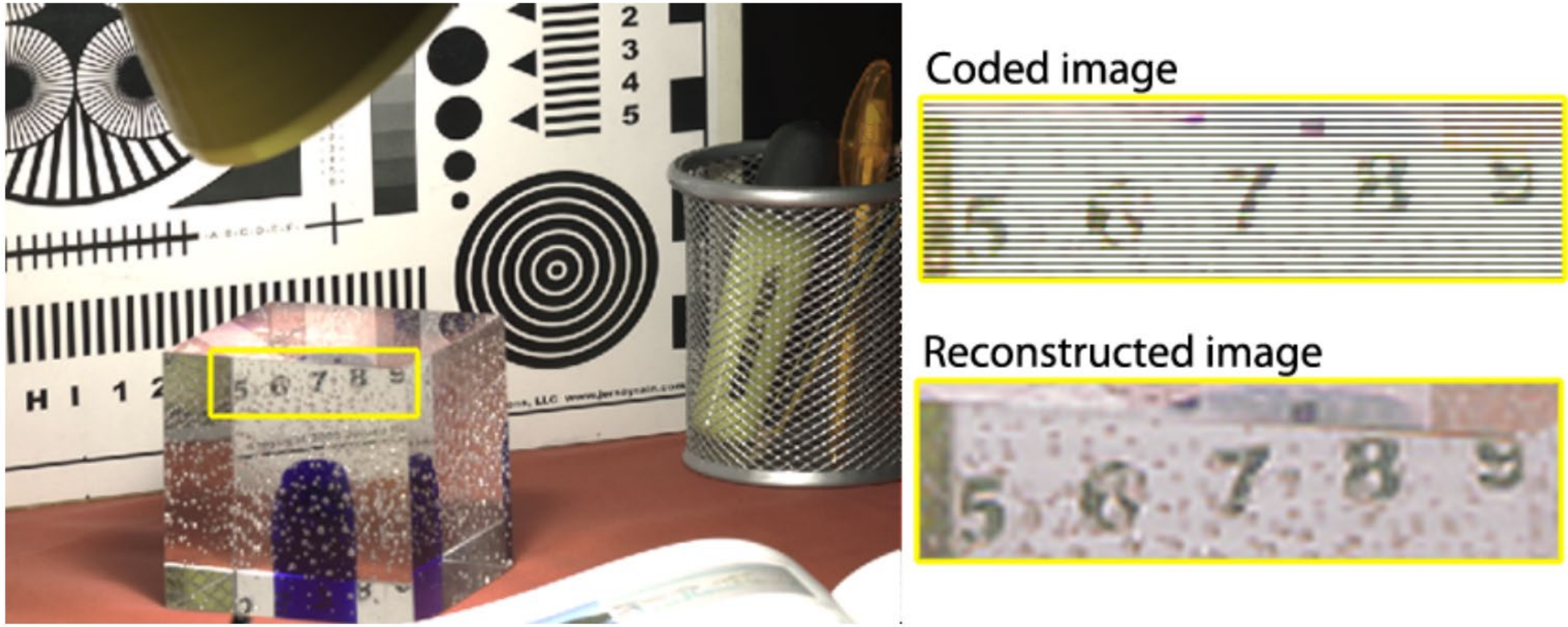
(b) Fused result

Mertens, Tom, Jan Kautz, and Frank Van Reeth. "Exposure Fusion." *Computer Graphics and Applications, Pacific Conference on*. IEEE Computer Society, 2007.

[Mertens 2007] http://ntp-0.cs.ucl.ac.uk/staff/j.kautz/publications/exposure_fusion.pdf

Dual ISO

- Two different exposures in the same shot
- Different ISO for different scanlines
- Readily available software (Magic Lantern: http://www.magiclantern.fm/)



Serrano, Ana, Felix Heide, Diego Gutierrez, Gordon Wetzstein, and Belen Masia. "Convolutional sparse coding for high dynamic range imaging." In *Computer Graphics Forum*, 2016.

[Serrano 2016] http://webdiis.unizar.es/~bmasia/pubs/eg2016/project_page_CSCHDR.html

Recap

- We’ve seen the problem of dynamic range
- We know how to capture HDR images
  - Using multi-bracketing, both with RAW or with non-linear images (needs radiometric calibration)
  - Other methods exist

Sources: CMU 15-463

The problem of dynamic range

- Our devices do not match the real world

- 10:1 photographic print (higher for glossy paper)
- 200:1 slide film
- 500:1 negative film
- 1,000:1 LCD display
- 2,000:1 digital SLR (at 12 bits)
- 100,000:1 real world

Two challenges:

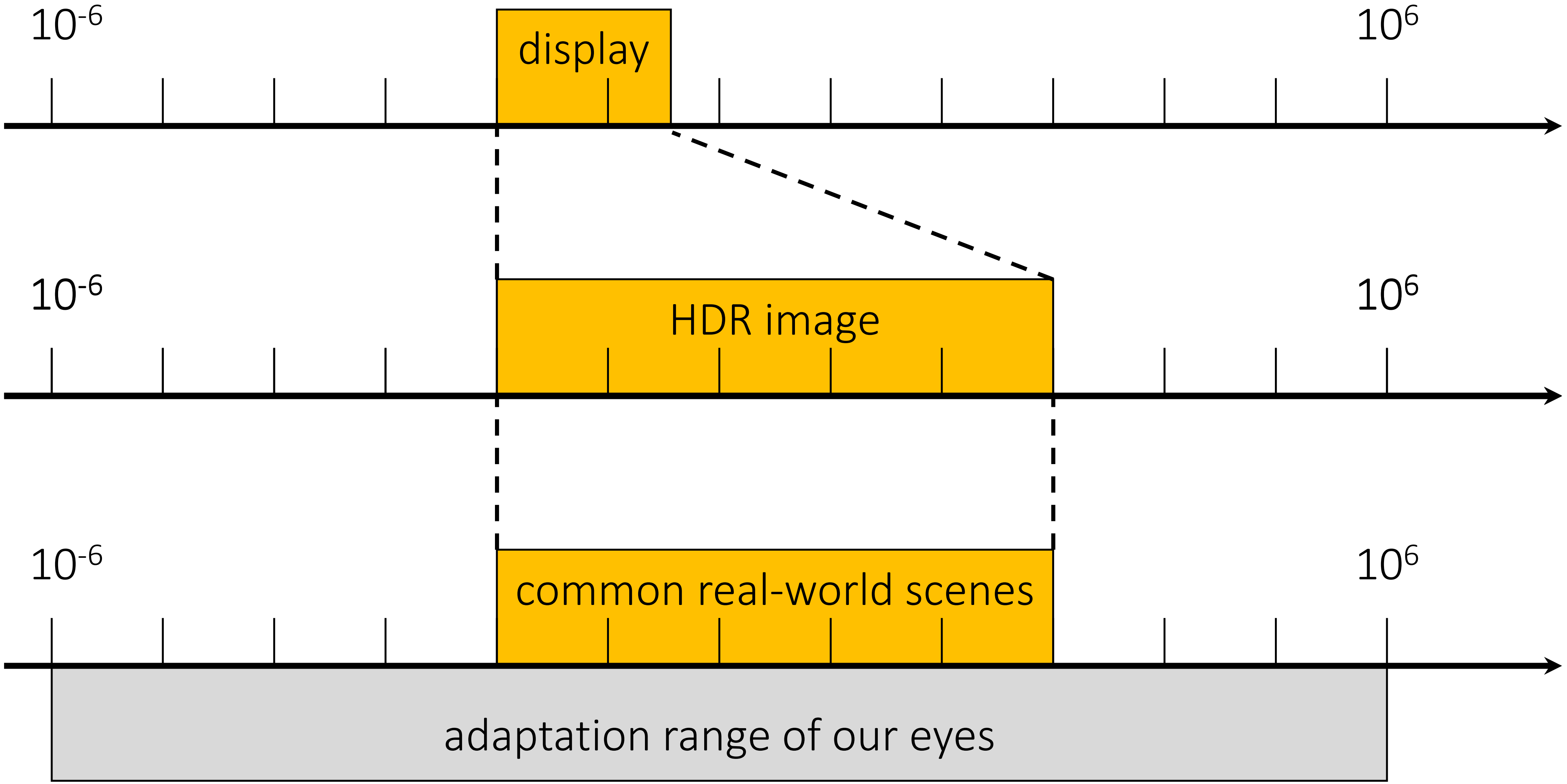
1. *HDR imaging: Which parts of the world do we include in the 8-14 bits available to our capture device?*
2. *Tonemapping: Which parts of the world do we display in the 4-10 bits available to our display device?*

Tonemapping

- Concept
- Basic techniques

Sources: CMU 15-463

How are HDR images displayed?



Sources: [CMU 15-463]

How are HDR images displayed?

The tone mapping (or tone reproduction) problem:

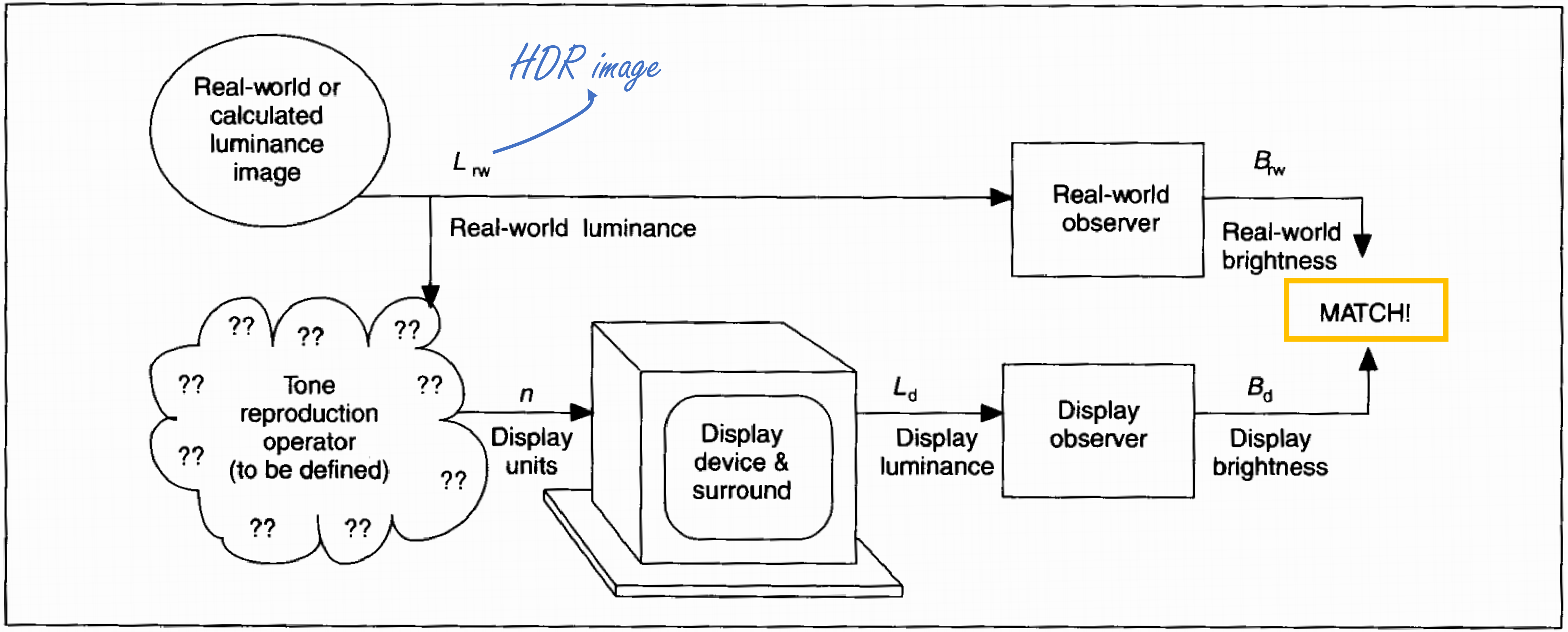
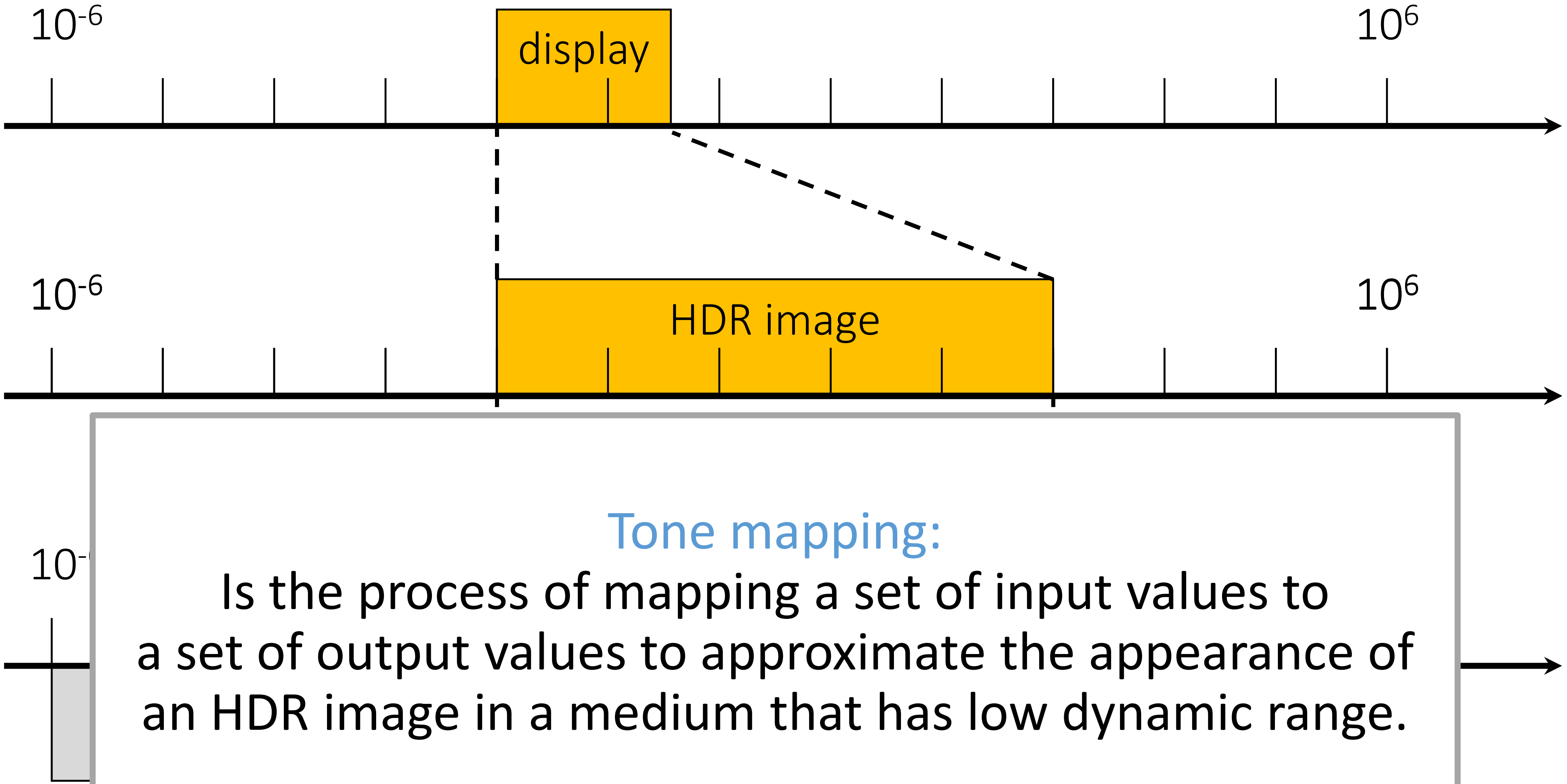


Figure 1. The tone reproduction problem: What operator will cause a close match between real-world and display brightness sensations?

Source: [Tumblin and Rushmeier 1993]

Sources: CMU 15-463

How are HDR images displayed?

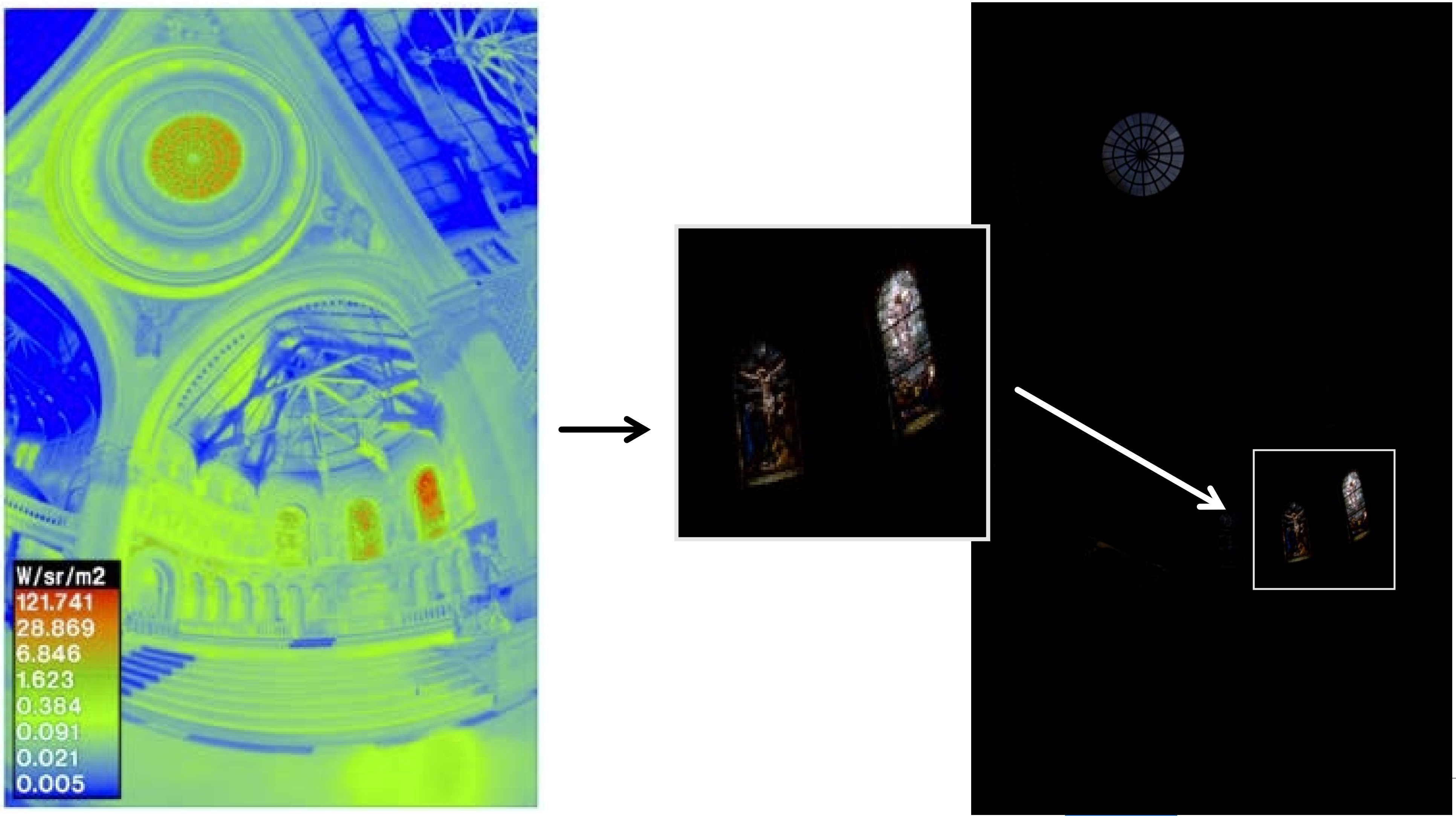


Tone mapping:

Is the process of mapping a set of input values to a set of output values to approximate the appearance of an HDR image in a medium that has low dynamic range.

Tonemapping: Linear scaling?

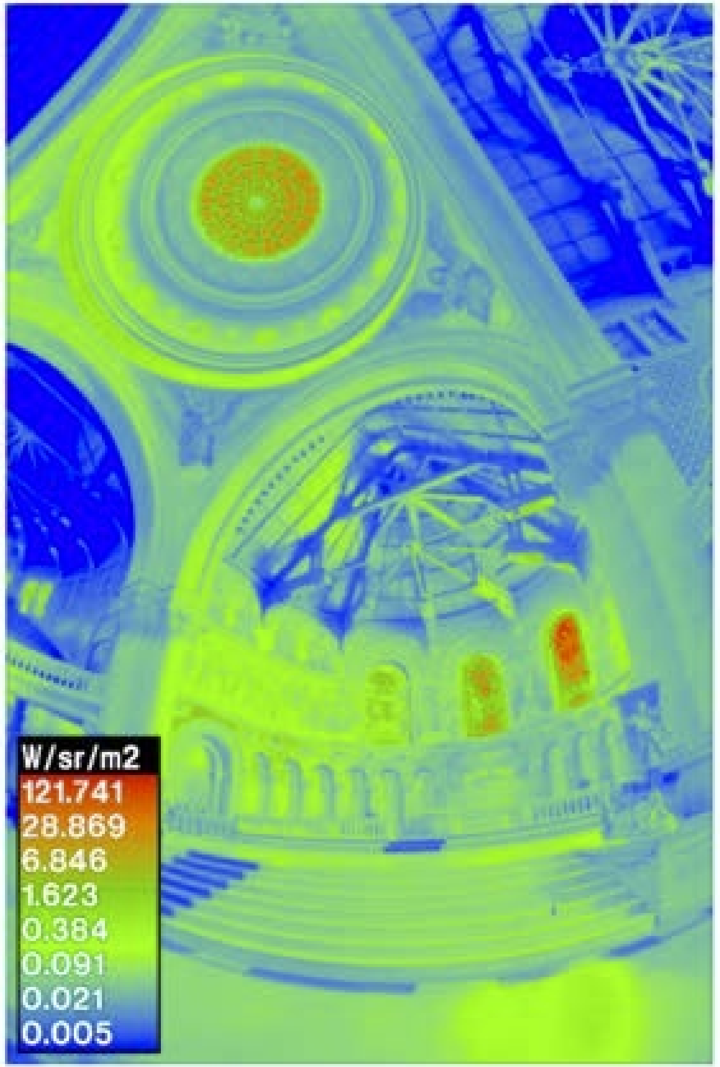
Scale image so that the maximum value is mapped to 1.

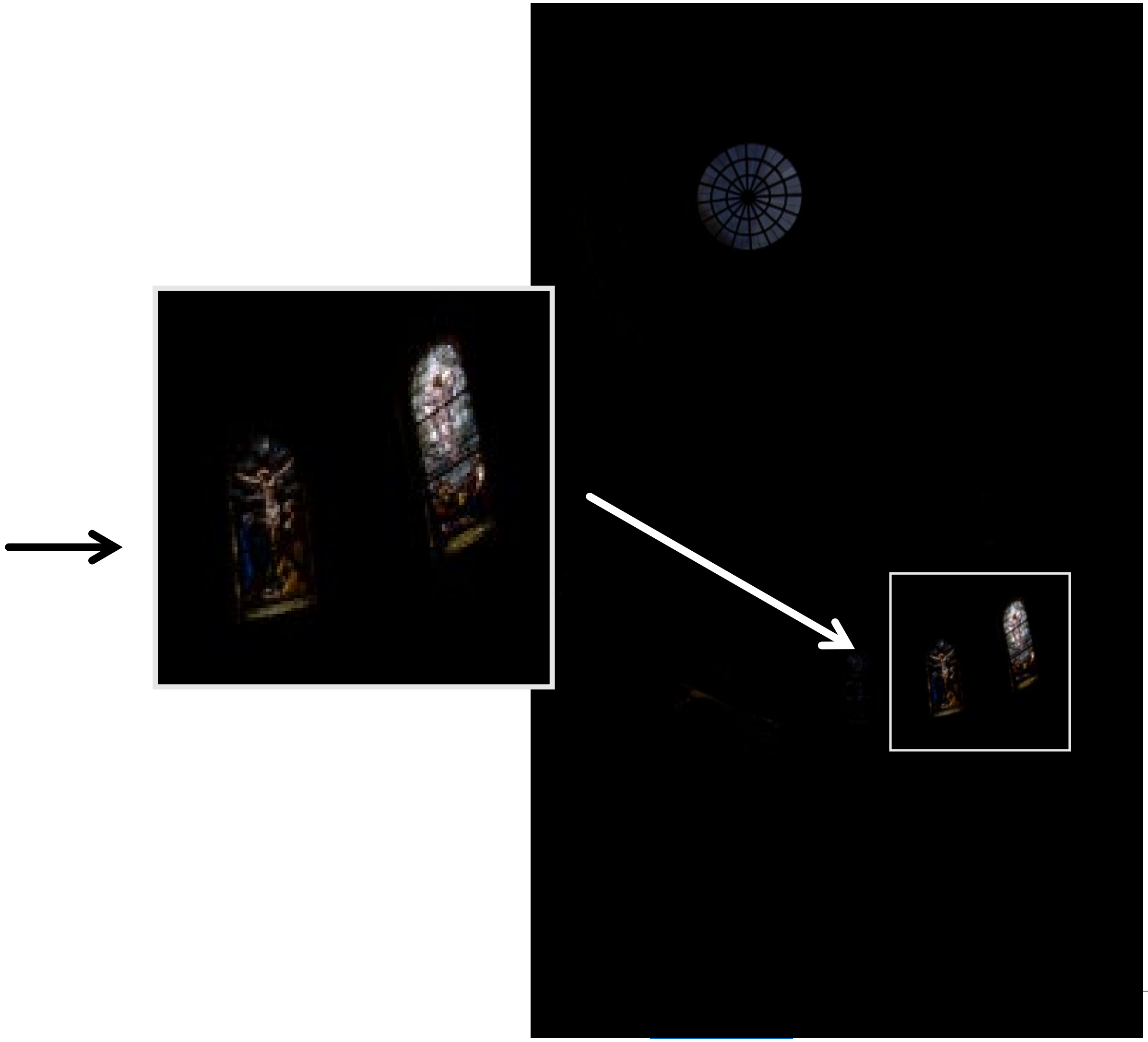


Tonemapping: Linear scaling?

Other ideas?

Scale image so that the maximum value is mapped to 1.

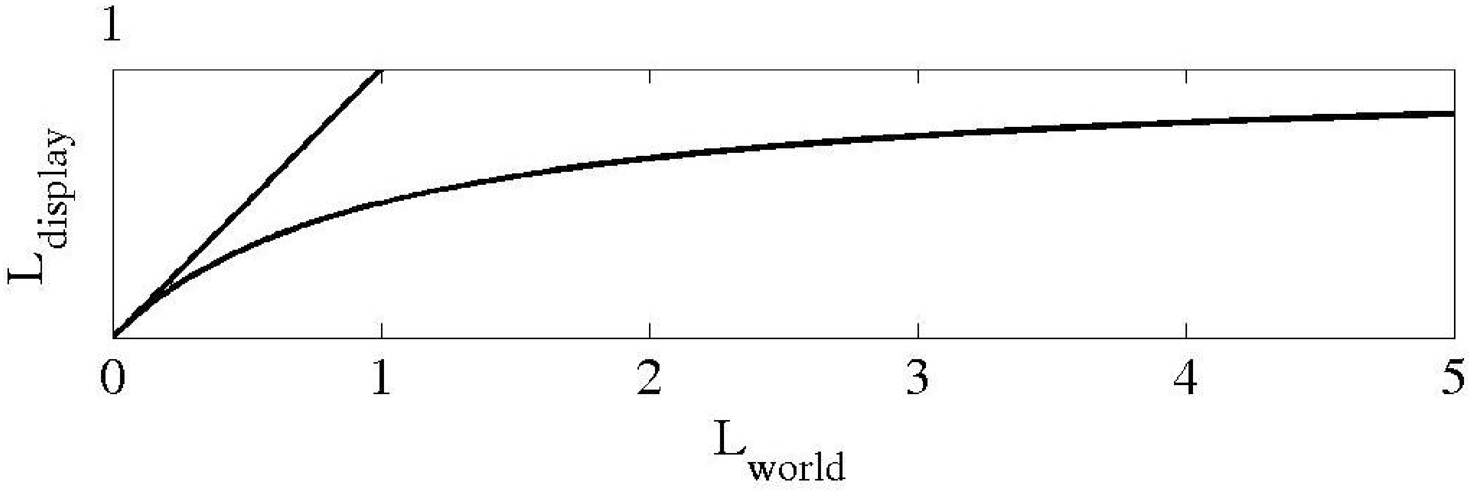




Photographic tonemapping

Apply the same non-linear scaling to all pixels in the image so that:

- Bring everything within range → asymptote to 1
- Leave dark areas alone → slope = 1 near 0

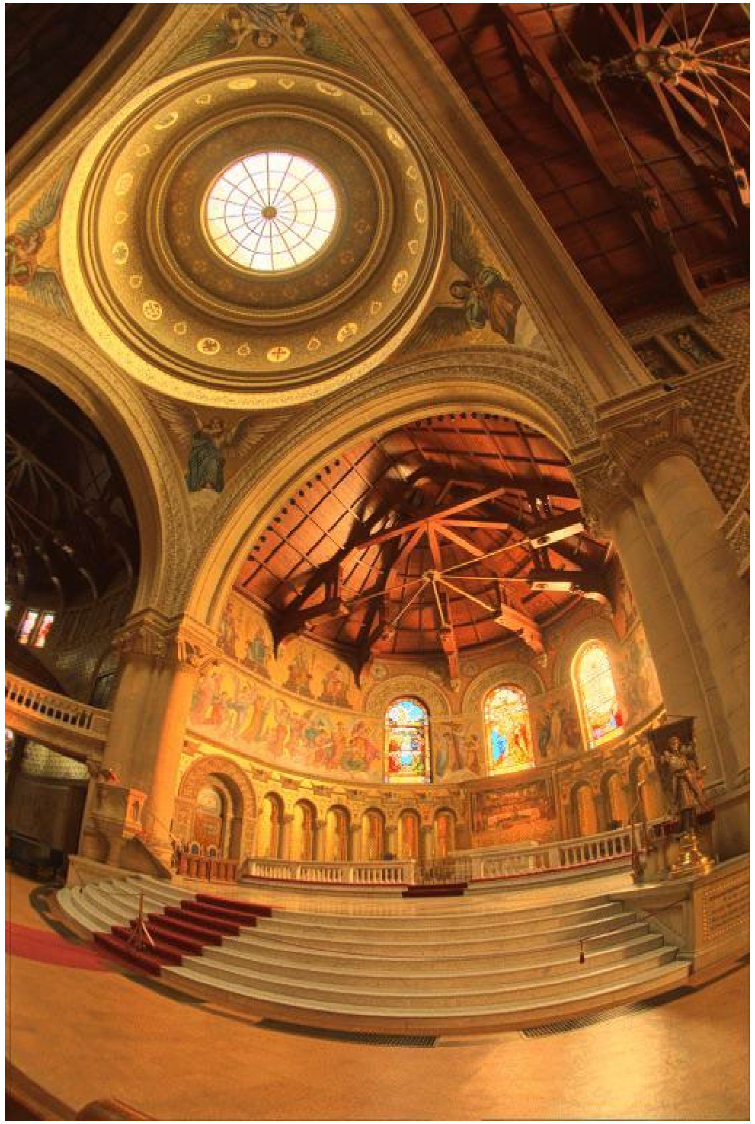


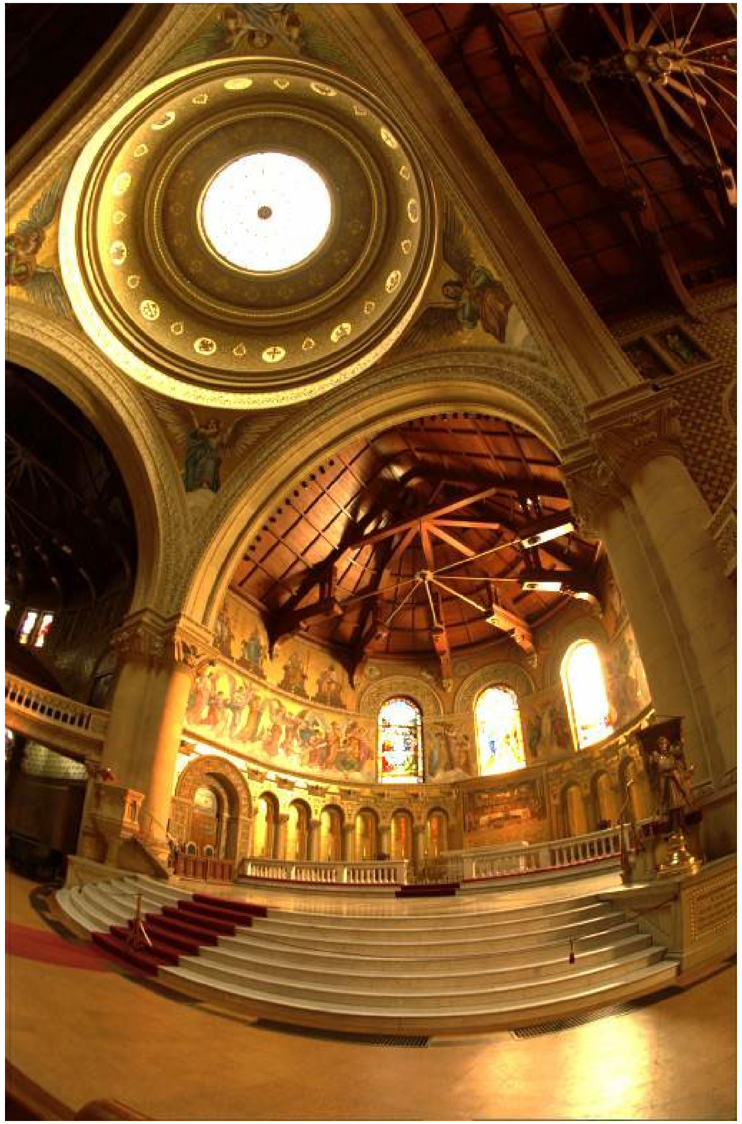
- Perceptually motivated, as it approximates our eye's response curve.

Sources: [CMU 15-463]

Photographic tonemapping: Results

photographic
tonemapping





linear scaling
(map 10% to 1)

Sources: CMU 15-463

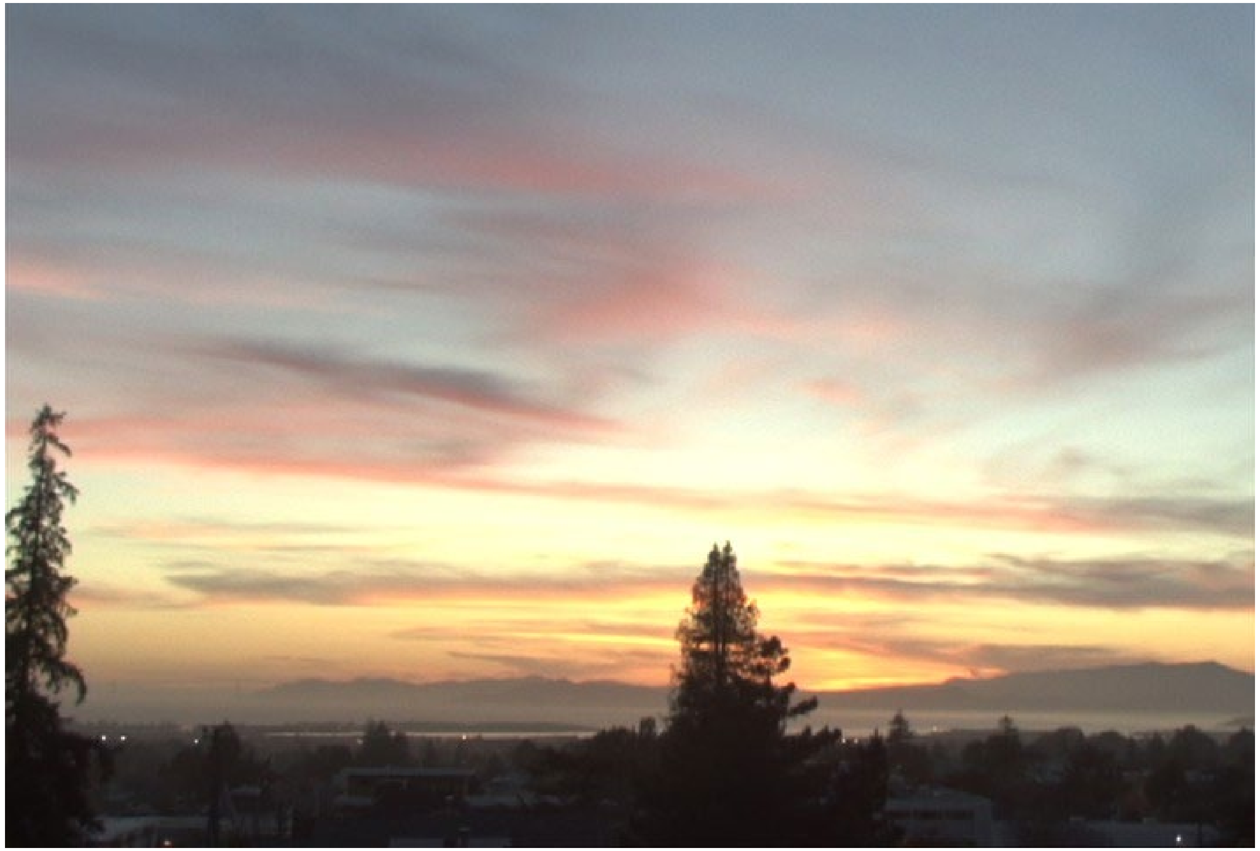
Photographic tonemapping: Comparison with LDR

Sources: [CMU 15-463]

Tonemapping: What happens to color?

How would you deal with this?

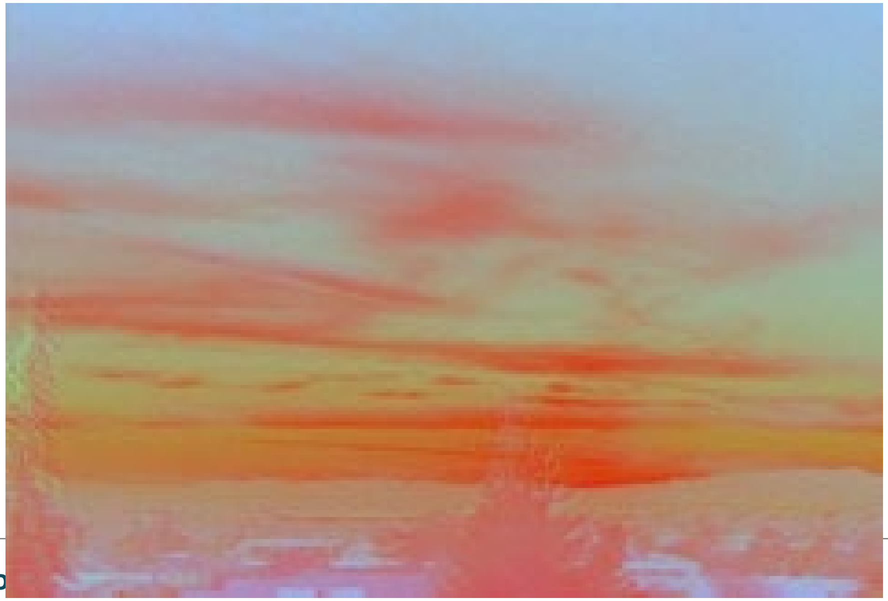
If we tonemap all channels the same, colors are washed out (RGB is not a perceptual color space)



Tonemapping: Color preservation

tonemap intensity

leave color the same

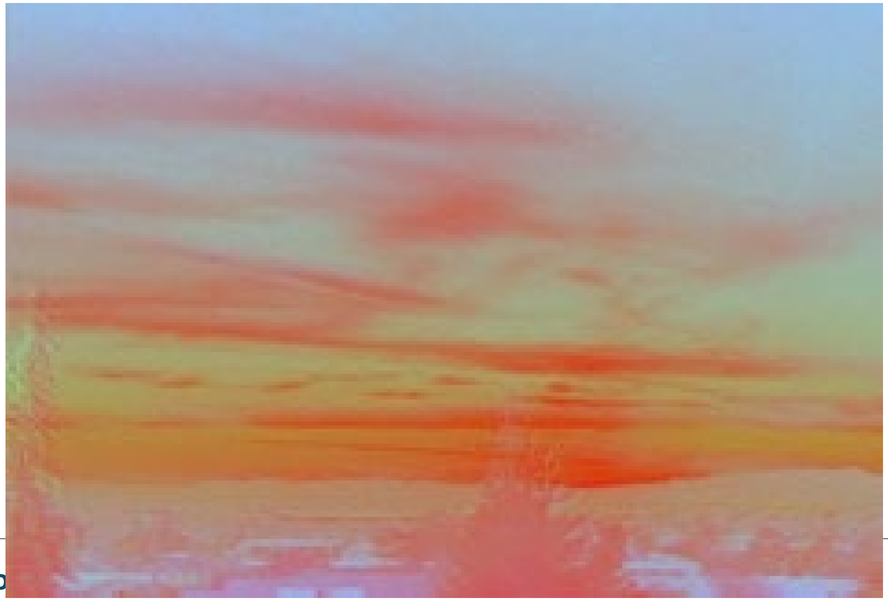


Sources: [CMU 15-463]

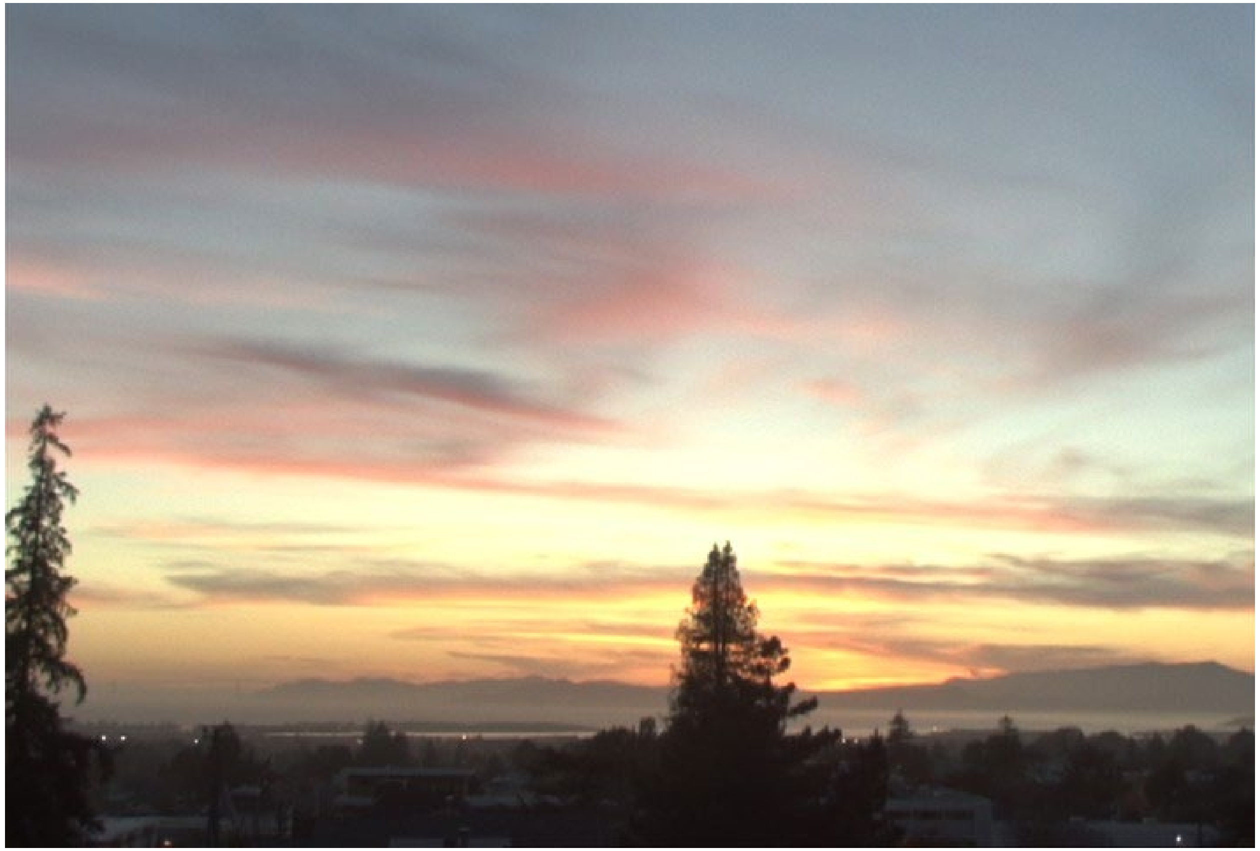
Tonemapping: Color preservation

tonemap intensity

leave color the same



Before

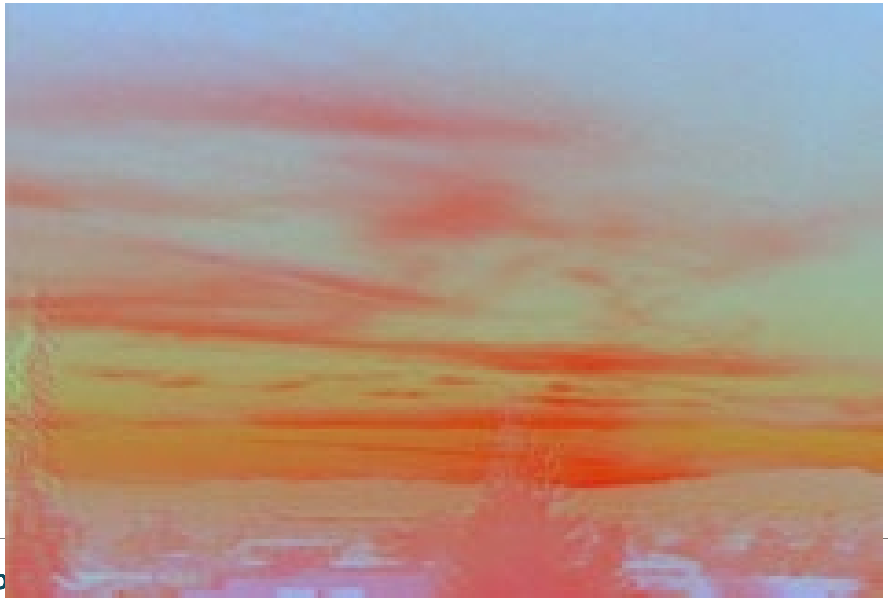


Tonemapping: Color preservation

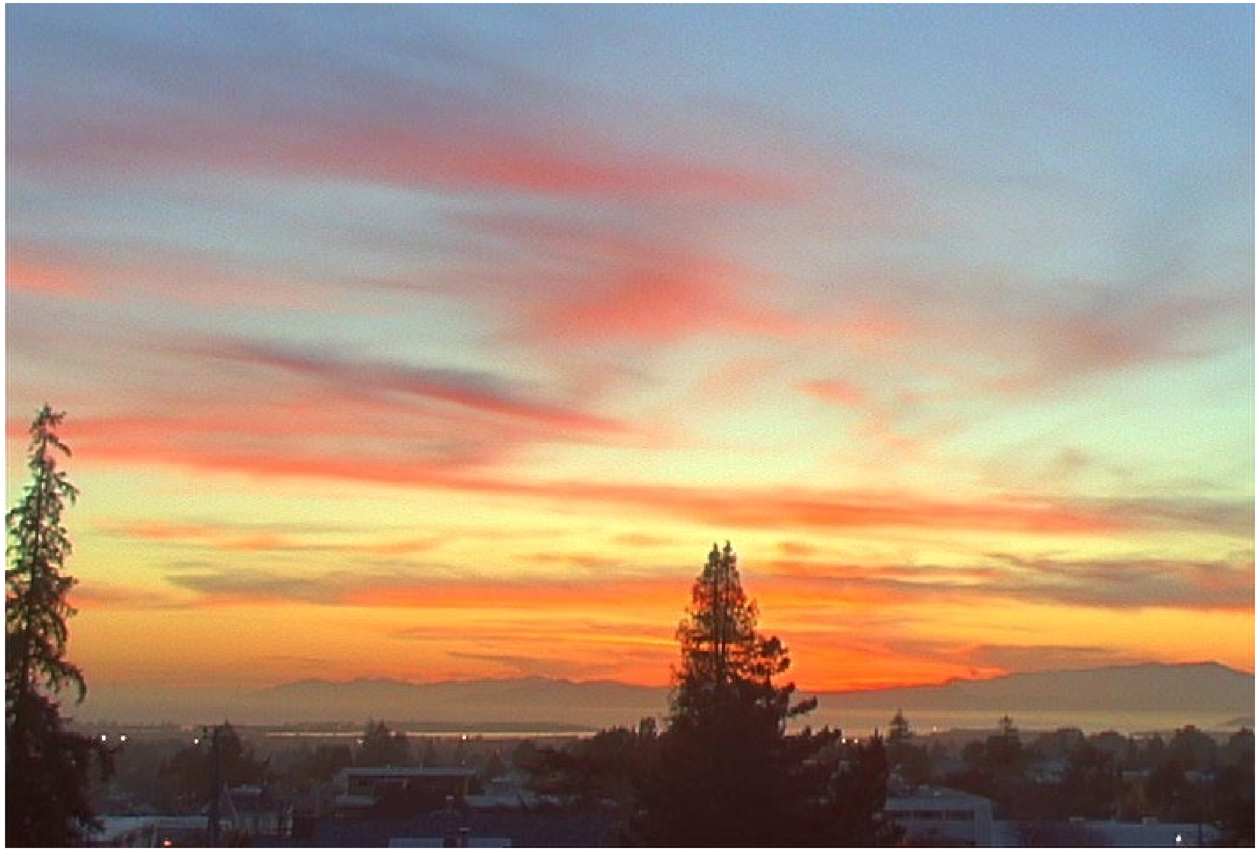
Implementation?

tonemap intensity

leave color the same



After



Tonemapping operators

- There are dozens of tonemapping algorithms (operators).
- Commercial SW tools implement some of them too, e.g., Adobe Photoshop

Exposure and Gamma	This method lets you manually adjust the exposure and gamma, which serve as the equivalent to brightness and contrast adjustment, respectively.
Highlight Compression	This method has no options and applies a custom tonal curve, which greatly reduces highlight contrast in order to brighten and restore contrast in the rest of the image.
Equalize Histogram	This method attempts to redistribute the HDR histogram into the contrast range of a normal 16 or 8-bit image. This uses a custom tonal curve which spreads out histogram peaks so that the histogram becomes more homogenous. It generally works best for image histograms which have several relatively narrow peaks with no pixels in between.
Local Adaptation	This is the most flexible method and probably the one which is of most use to photographers. Unlike the other three methods, this one changes how much it brightens or darkens regions on a per-pixel basis (similar to local contrast enhancement). This has the effect of tricking the eye into thinking that the image has more contrast, which is often critical in contrast-deprived HDR images. This method also allows changing the tonal curve to better suit the image.

Sources: CMU 15-463

Tonemapping operators

Chao Wang, Bin Chen, Hans-Peter Seidel, Karol Myszkowski, and Ana Serrano. "Learning a self-supervised tone mapping operator via feature contrast masking loss." In *Computer Graphics Forum* (2022).

- An active area of research

Recap

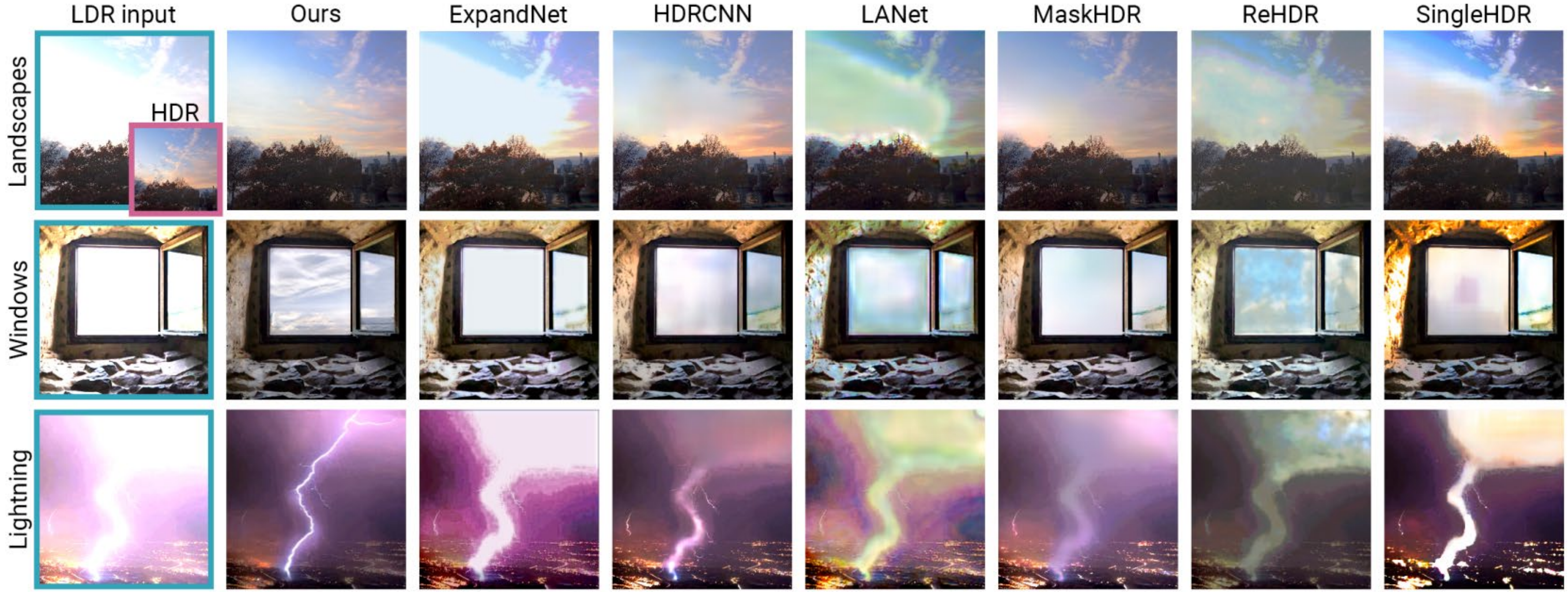
- We've seen the problem of dynamic range
- We know how to capture HDR images
  - Using multi-bracketing, both with RAW or with non-linear images (needs radiometric calibration)
  - Other methods exist
- We know how to display HDR images (tonemapping)

Sources: [CMU 15-463]

Inverse tone mapping

Chao Wang, Ana Serrano, Xingang Pan, Bin Chen, Hans-Peter Seidel, Christian Theobalt, Karol Myszkowski, and Thomas Leimkuehler. "GlowGAN: Unsupervised Learning of HDR Images from LDR Images in the Wild". 2023.

- The process of recovering HDR images from LDR content
- Useful for extending legacy (LDR) content to HDR
- An active area of research



Some final notes about HDR imaging and tonemapping

Sources: [CMU 15-463]

A note about terminology

"High-dynamic-range imaging" is used to refer to a lot of different things:

1. Using single RAW images.

2. Performing radiometric calibration.

3. Merging an exposure stack.

4. Tonemapping an image (linear or non-linear, HDR or LDR).

5. Some or all of the above.

Technically, HDR imaging is simply the process of creating a radiometrically linear image, free of overexposure and underexposure artifacts. This is achieved using some combination of 1-3, depending on the imaging scenario.

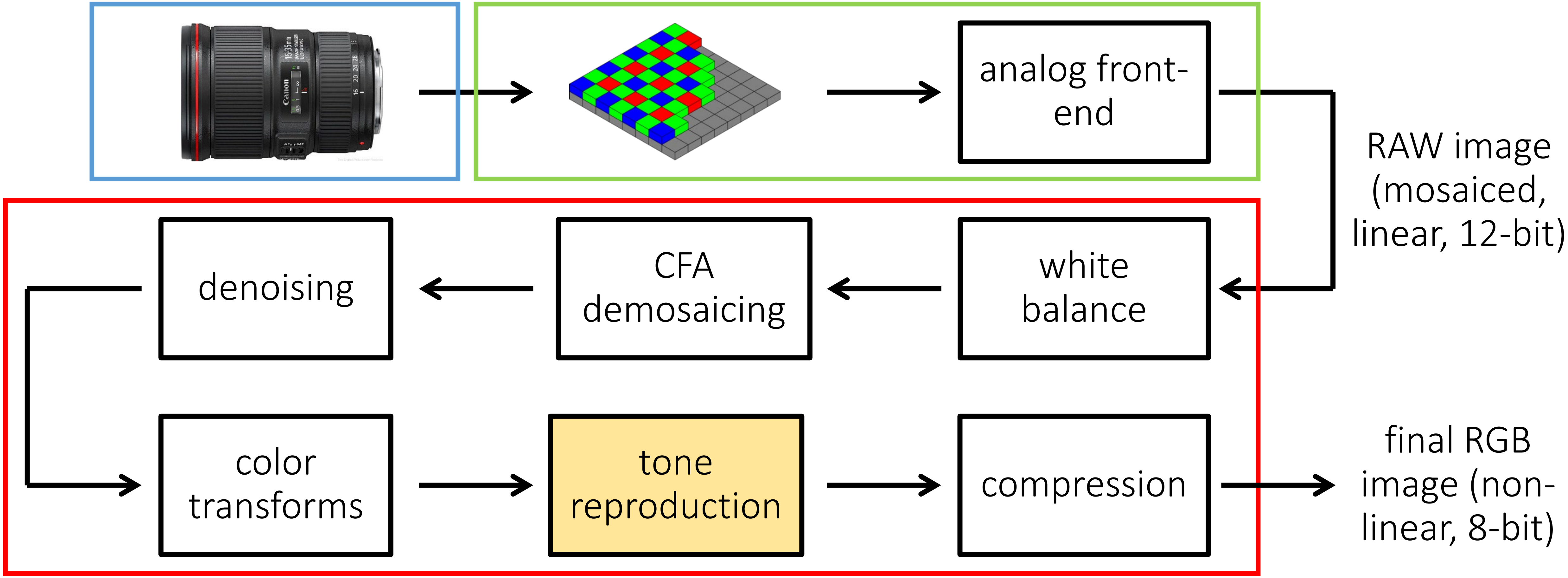
In consumer photography, HDR photography includes step 4 (tonemapping).

Sources: [CMU 15-463]

Another note about terminology

Tonemapping is just another form of tone reproduction.

- Many ISPs implement the tonemapping algorithms we discussed for tone reproduction.



Sources: [CMU 15-463]

A note of caution

- HDR photography can produce very visually compelling results.

* It also has other applications -> it can store the absolute radiance values of a scene.

Sources: [CMU 15-463]

A note of caution

- HDR photography can produce very visually compelling results.

- It is also a very routinely abused technique, resulting in bad results.

Sources: [CMU 15-463]

A note of caution

- HDR photography can produce very visually compelling results.
- It is also a very routinely abused technique, resulting in bad results.
- The problem typically is tonemapping, not HDR imaging itself.

Sources: [CMU 15-463]

A note about HDR today

- Most cameras (including phone cameras) have automatic HDR modes/apps.
- It is a popular-enough feature that phone manufacturers are actively competing about which one has the best HDR.
- The technology behind some of those apps (e.g., Google's HDR+) is published in SIGGRAPH and SIGGRAPH Asia conferences.

Burst photography for high dynamic range and low-light imaging on mobile cameras

Samuel W. Hasinoff, Dillon Sharlet, Ryan Geiss, Andrew Adams, Jonathan T. Barron, Florian Kainz, Jiawen Chen, Marc Levoy

Google Research

Figure 1: *A comparison of a conventional camera pipeline (left, middle) and our burst photography pipeline (right) running on the same cell-phone camera. In this low-light setting (about 0.7 lux), the conventional camera pipeline underexposes (left). Brightening the image (middle) reveals heavy spatial denoising, which results in loss of detail and an unpleasantly blotchy appearance. Fusing a burst of images increases the signal-to-noise ratio, making aggressive spatial denoising unnecessary. We encourage the reader to zoom in. While our pipeline excels in low-light and high-dynamic-range scenes (for an example of the latter see figure 10), it is computationally efficient and reliably artifact-free, so it can be deployed on a mobile camera and used as a substitute for the conventional pipeline in almost all circumstances. For readability the figure has been made uniformly brighter than the original photographs.*

Abstract

Cell phone cameras have small apertures, which limits the number of photons they can gather, leading to noisy images in low light. They also have small sensor pixels, which limits the number of electrons each pixel can store, leading to limited dynamic range. We describe a computational photography pipeline that captures, aligns, and merges a burst of frames to reduce noise and increase dynamic range. Our system has several key features that help make it robust and efficient. First, we do not use bracketed exposures. Instead, we capture frames of constant exposure, which makes alignment more robust, and we set this exposure low enough to avoid blowing out highlights. The resulting merged image has clean shadows and high bit depth, allowing us to apply standard HDR tone mapping methods. Second, we begin from Bayer raw frames rather than the demosaicked RGB (or YUV) frames produced by hardware Image Signal Processors (ISPs) common on mobile platforms. This gives us more bits per pixel and allows us to circumvent the ISP's unwanted tone mapping and spatial denoising. Third, we use a novel FFT-based alignment algorithm and a hybrid 2D/3D Wiener filter to denoise and merge the frames in a burst. Our implementation is built atop Android's Camera2 API, which provides per-frame camera control and access to raw imagery, and is written in the Halide domain-specific language (DSL). It runs in 4 seconds on device (for a 12 Mpix image), requires no user intervention, and ships on several mass-produced cell phones.

Keywords: computational photography, high dynamic range

Concepts: **•Computing methodologies → Computational photography; Image processing;**

1 Introduction

The main technical impediment to better photographs is lack of light. In indoor or night-time shots, the scene as a whole may provide insufficient light. The standard solution is either to apply analog or digital gain, which amplifies noise, or to lengthen exposure time, which causes motion blur due to camera shake or subject motion. Surprisingly, daytime shots with high dynamic range may also suffer from lack of light. In particular, if exposure time is reduced to avoid

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Take-home messages

- Real scenes have a high dynamic range (HDR), and our eyes are able to adapt to them.
  Conventional cameras, displays, and images support low dynamic range (LDR).
- HDR images store intensity values as floating point 32-bit or 64-bit values.
- HDR images can be captured with conventional cameras using multi-bracketing.
- HDR images can be displayed in conventional displays using tonemapping.

References

Basic reading:

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- Mitsunaga and Nayar, "Radiometric self calibration," CVPR 1999.

    The two classical papers on radiometric calibration and HDR imaging, which more or less started HDR imaging research in computer vision and graphics.
- Reinhard et al., "Photographic Tone Reproduction for Digital Images," SIGGRAPH 2002.

    The photographic tonemapping paper, including a very nice discussion of the zone system for film.

Additional reading:

- Reinhard et al., "High Dynamic Range Imaging, Second Edition: Acquisition, Display, and Image-Based Lighting," Morgan Kaufmann 2010.

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- Durand and Dorsey, "Fast bilateral filtering for the display of high-dynamic-range images," SIGGRAPH 2002.

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    The paper describing Google's HDR+.
- Ward, "The radiance lighting simulation and rendering system," SIGGRAPH 1994.
- Ward, "High Dynamic Range Image Encodings," http://www.anyhere.com/gward/hdrenc/hdr_encodings.html

    The paper that introduced (among other things) the .hdr image format for HDR images. The website has a very detailed discussion of HDR image formats.
- Kuang et al., "Evaluating HDR rendering algorithms," TAP 2007.

    One of many, many papers trying to do a perceptual evaluation of different tonemapping algorithms.
- Levoy, "Extreme imaging using cell phones," http://graphics.stanford.edu/talks/seeinthedark-public-15sep16.key.pdf

    A set of slides by Marc Levoy on the challenges of HDR imaging and modern approaches for addressing them.